Theory of Mathematics

Volume I: Linear Algebra

Contents

Part I

Vector Spaces

Chapter 1

Scalars, Vectors and Linear Combinations

Linear algebra begins with two operations: adding vectors and scaling them by elements of a field. The point of the abstract definition is not to hide geometry but to identify exactly which arguments depend only on those operations. This chapter moves back and forth between concrete coordinate vectors, functions and polynomials so that linear combinations become a working language rather than a formal slogan.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use fields and scalars in basis-free and coordinate form;
- explain and use vector spaces in basis-free and coordinate form;
- explain and use linear combinations in basis-free and coordinate form;
- explain and use concrete models in basis-free and coordinate form;
- explain and use affine versus linear geometry in basis-free and coordinate form;
- explain and use complex scalars and realification in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

1.1 Fields and scalars

We work over a field $\mathbb{F}$, usually $\mathbb{R}$ or $\mathbb{C}$. The field laws allow addition, subtraction, multiplication and division by nonzero scalars. Every vector-space identity ultimately rests on these laws together with the vector-space axioms.

1.2 Vector spaces

A vector space V over $\mathbb{F}$ is an abelian group under addition equipped with scalar multiplication $\mathbb{F} \times V \to V$ satisfying distributivity, compatibility, and $1v = v$. The zero vector is unique, and $0v = 0$, $a0 = 0$, and $(-1)v = -v$ follow from the axioms.

1.3 Linear combinations

Given $v_1, \ldots, v_r \in V$, a linear combination is $a_1v_1 + \cdots + a_rv_r$ with $a_i \in \mathbb{F}$. Finite linear combinations are the algebraic mechanism for building new vectors from old ones.

1.4 Concrete models

The spaces $\mathbb{F}^n$, polynomial spaces $\mathbb{F}[x]$, matrix spaces $M_{m,n}(\mathbb{F})$, and function spaces $\mathbb{F}^X$ all satisfy the same vector-space axioms. Coordinatewise formulas are examples of a common structure.

1.5 Affine versus linear geometry

Linear combinations with arbitrary coefficients preserve the origin. Affine combinations impose $a_1 + \cdots + a_r = 1$ and describe points in affine hulls. Convex combinations add $a_i \geq 0$ and describe convex hulls.

1.6 Complex scalars and realification

A complex vector space is automatically a real vector space after restricting scalars, but its real dimension doubles when the complex dimension is finite. Forgetting part of the scalar field changes the linear relations that are allowed.

1.7 Core structural results

Theorem 1.1 (Uniqueness consequences of the axioms). *For every vector space V, the zero vector and additive inverse of each vector are unique; moreover $0v = 0$, $a0 = 0$, and $(-a)v = -(av)$ for all $a \in \mathbb{F}$.*

Proof. Each identity follows by adding a suitable additive inverse and using distributivity. For example, $0v = (0 + 0)v = 0v + 0v$; cancelling $0v$ gives $0v = 0$. □

Theorem 1.2 (Linear-combination regrouping). *A linear combination of linear combinations is again a linear combination of the original vectors.*

Proof. Substitute each inner expression and use associativity and distributivity to collect the coefficient of each original vector. □

1.8 Worked examples

Example 1.3 (Polynomials). In $\mathbb{R}[x]$, the polynomial $3 - 2x + 5x^3$ is a linear combination of $1, x, x^3$. The same expression may be viewed inside any larger polynomial space without changing its coefficients.

Example 1.4 (Functions)**.** If $f, g : X \to \mathbb{R}$, then $2f - 3g$ is the function $x \mapsto 2f(x) - 3g(x)$. Function spaces make clear that vectors need not be arrows.

Example 1.5 (Matrices)**.** The matrix $\begin{pmatrix} 1 & 2 \\ 0 & -1 \end{pmatrix}$ is a linear combination of the four matrix units E_{ij}. Matrix units play the role of coordinate directions.

1.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

1.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 1.6 (A hidden dependence in three vectors)**.** In $\mathbb{R}^3$ let $u = (1, 1, 0)$, $v = (0, 1, 1)$ and $w = (1, 2, 1)$. Determine all triples (a, b, c) such that $au + bv + cw = 0$ and interpret the answer.

Solution. Since $w = u + v$, the relation is $(a + c)u + (b + c)v = 0$. The vectors u, v are independent, so $a + c = b + c = 0$. Hence $(a, b, c) = t(-1, -1, 1)$. The one-dimensional relation space records exactly one redundancy: w contributes no new direction. □

Problem 1.7 (Changing the scalar field)**.** Regard $\mathbb{C}$ first as a vector space over $\mathbb{C}$ and then over $\mathbb{R}$. Compare the linear combination $i \cdot 1$ in the two structures and explain why the dimensions differ.

Solution. Over $\mathbb{C}$, multiplication by i is scalar multiplication, so $i = i \cdot 1$ and $\{1\}$ is a basis. Over $\mathbb{R}$, i is not an available scalar; one needs both 1 and i, and every $z = a + bi$ is the real linear combination $a \cdot 1 + b \cdot i$. Thus the dimensions are 1 over $\mathbb{C}$ and 2 over $\mathbb{R}$. □

Problem 1.8 (Polynomial coordinates without a basis theorem)**.** Express $p(x) = 2 - 3x + 5x^2$ as a linear combination of $q_1 = 1 + x$, $q_2 = x + x^2$, and $q_3 = 1 - x^2$.

Solution. Seek $p = aq_1 + bq_2 + cq_3$. Comparing coefficients gives $a + c = 2$, $a + b = -3$, and $b - c = 5$. The third equation follows from the first two, so there are infinitely many representations: taking $a = t$ gives $b = -3 - t$ and $c = 2 - t$. This nonuniqueness reveals that q_1, q_2, q_3 are dependent. □

Problem 1.9 (Affine but not linear)**.** Show that the line $L = \{(x, y) : x + y = 1\}$ is closed under affine combinations but is not a vector subspace of $\mathbb{R}^2$.

Solution. If $p, q \in L$ and $\alpha + \beta = 1$, then the coordinate sum of $\alpha p + \beta q$ is $\alpha \cdot 1 + \beta \cdot 1 = 1$, so L is affine-combination closed. But $(0, 0) \notin L$, so L cannot be a vector subspace. □

Problem 1.10 (Matrix units as building blocks)**.** Write $A = \begin{pmatrix} 2 & -1 \\ 3 & 4 \end{pmatrix}$ as a linear combination of the matrix units $E_{11}, E_{12}, E_{21}, E_{22}$ and explain why the representation is unique.

Solution. $A = 2E_{11} - E_{12} + 3E_{21} + 4E_{22}$. If a linear combination of the four matrix units is zero, inspection of the four entries forces every coefficient to vanish. Thus the units are independent, and entrywise comparison also shows they span $M_2(\mathbb{R})$. ☐

Problem 1.11 (Functions as vectors)**.** Let $f(t) = 1$, $g(t) = t$, and $h(t) = 1 + t$ in the vector space of real functions on $[0, 1]$. Find every relation $af + bg + ch = 0$.

Solution. The identity means $(a + c) + (b + c)t = 0$ for every t. Hence $a + c = 0$ and $b + c = 0$, so $(a, b, c) = s(-1, -1, 1)$. Again $h = f + g$, illustrating that dependence is structural and not tied to coordinate vectors. ☐

Problem 1.12 (Convex versus affine combinations)**.** Can $(3, 0)$ be written as a convex combination of $(0, 0)$ and $(2, 0)$? Can it be written as an affine combination?

Solution. A convex combination has the form $(2\beta, 0)$ with $\alpha + \beta = 1$ and $\alpha, \beta \geq 0$, so $0 \leq 2\beta \leq 2$; $(3, 0)$ is impossible. For an affine combination, nonnegativity is dropped: choose $\beta = 3/2$ and $\alpha = -1/2$. ☐

Problem 1.13 (Complex linearity test)**.** Define $T : \mathbb{C} \to \mathbb{C}$ by $T(z) = \overline{z}$. Is T linear over $\mathbb{R}$? Over $\mathbb{C}$?

Solution. For real scalars a, b, conjugation satisfies $T(az + bw) = aT(z) + bT(w)$, so it is real-linear. But $T(iz) = -i\overline{z}$ while $iT(z) = i\overline{z}$, so it is not complex-linear unless $z = 0$. ☐

Problem 1.14 (A parameterized linear combination)**.** For which real t does $(1, t, t^2)$ belong to the span of $(1, 0, 1)$ and $(0, 1, 1)$?

Solution. A general span vector is $(a, b, a + b)$. Matching gives $a = 1$, $b = t$, and therefore $t^2 = 1 + t$. Thus t must solve $t^2 - t - 1 = 0$, so $t = (1 \pm \sqrt{5})/2$. ☐

Problem 1.15 (Why finite sums are enough)**.** Explain why the definition of span uses finite rather than arbitrary infinite linear combinations, even in an infinite-dimensional vector space.

Solution. Vector-space axioms define only binary addition and its finite iteration. No notion of convergence or infinite summation is part of a general vector space. Infinite sums require extra topological or analytic structure, so algebraic span is necessarily built from finite combinations. ☐

Problem 1.16 (Direct verification from axioms)**.** Prove from the vector-space axioms that if $av = 0$ and $a \neq 0$, then $v = 0$.

Solution. Multiply the equation by a^{-1}. Compatibility of scalar multiplication gives $v = 1v = (a^{-1}a)v = a^{-1}(av) = a^{-1}0 = 0$. This cancellation principle is used constantly in linear algebra. ☐

Problem 1.17 (A geometric synthesis)**.** Describe the set of all linear combinations of two nonparallel vectors in $\mathbb{R}^3$ and compare it with their affine hull.

Solution. The linear combinations form the plane through the origin spanned by the two vectors. Their affine hull as two points consists only of the line through those two points. The distinction is that span treats the vectors as directions from the origin, whereas affine hull treats them as points with coefficients summing to one. ☐

1.11 Exercises with complete solutions

Exercise 1.18. Show directly from the axioms that $a(-v) = -(av)$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. Use $v + (-v) = 0$, multiply by a, and use distributivity. The resulting equation $av + a(-v) = 0$ says that $a(-v)$ is the additive inverse of av. $\square$

Exercise 1.19. Write $(2, -1, 3)$ as a linear combination of the standard vectors of $\mathbb{R}^3$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. $(2, -1, 3) = 2e_1 - e_2 + 3e_3$. $\square$

Exercise 1.20. Give a vector-space structure on the set of real-valued continuous functions on $[0, 1]$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. Use pointwise addition and scalar multiplication. Continuity is preserved by both operations, so the operations remain inside the set. $\square$

Exercise 1.21. Why is the unit circle in $\mathbb{R}^2$ not a vector space?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. It does not contain the zero vector and is not closed under scalar multiplication; for example $2(1, 0) = (2, 0)$ is not on the unit circle. $\square$

Exercise 1.22. Over which scalars is $\mathbb{C}$ naturally a one-dimensional vector space? What is its dimension over $\mathbb{R}$?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. It is one-dimensional over $\mathbb{C}$, with basis $\{1\}$, and two-dimensional over $\mathbb{R}$, with basis $\{1, i\}$. $\square$

Exercise 1.23. Find an affine combination of $(0, 0)$ and $(2, 0)$ giving $(3, 0)$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. Take coefficients $-\frac{1}{2}$ and $\frac{3}{2}$; they sum to 1 and give $(3, 0)$. $\square$

Exercise 1.24. Show that the set of all 2×2 real matrices is a vector space.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. Matrix addition is an abelian group operation and scalar multiplication satisfies the vector-space laws entrywise. $\square$

Exercise 1.25. Explain why a finite sum $\sum_i a_i v_i$ is independent of parentheses.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Associativity of vector addition allows any parenthesization; a formal induction on the number of summands proves the claim. □

Chapter summary

Linear algebra begins with two operations: adding vectors and scaling them by elements of a field. The point of the abstract definition is not to hide geometry but to identify exactly which arguments depend only on those operations. This chapter moves back and forth between concrete coordinate vectors, functions and polynomials so that linear combinations become a working language rather than a formal slogan. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 2

Subspaces, Span and Linear Independence

Subspaces are the linear subsets of a vector space, span is the smallest subspace generated by specified vectors, and linear independence is the condition that a generating description has no redundancy. These three ideas form the basic grammar of finite-dimensional reasoning.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use subspaces in basis-free and coordinate form;
- explain and use intersections and sums in basis-free and coordinate form;
- explain and use span in basis-free and coordinate form;
- explain and use linear dependence in basis-free and coordinate form;
- explain and use redundancy and exchange in basis-free and coordinate form;
- explain and use homogeneous systems in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

2.1 Subspaces

A subset $W \subseteq V$ is a subspace when it contains 0 and is closed under addition and scalar multiplication. Equivalently, $au + bv \in W$ for all $u, v \in W$ and $a, b \in \mathbb{F}$.

2.2 Intersections and sums

Arbitrary intersections of subspaces are subspaces. The union of two subspaces need not be a subspace, but the sum $U + W = \{u + w\}$ always is.

2.3 Span

The span of $S \subseteq V$ is the set of all finite linear combinations of elements of S. It is the smallest subspace containing S.

2.4 Linear dependence

A finite list $v_1, \dots, v_r$ is linearly dependent when some nontrivial scalar relation $a_1v_1 + \cdots + a_rv_r = 0$ exists. Otherwise the list is linearly independent.

2.5 Redundancy and exchange

If a list is dependent, at least one vector is in the span of the others. This simple observation drives the exchange arguments used to prove uniqueness of dimension.

2.6 Homogeneous systems

Columns of a matrix are linearly dependent exactly when the corresponding homogeneous system has a nonzero solution. Row reduction converts an abstract dependence question into an algorithm.

2.7 Core structural results

Theorem 2.1 (Subspace criterion)**.** *A nonempty subset $W \subseteq V$ is a subspace if and only if $au + bv \in W$ for all $u, v \in W$ and all scalars a, b.*

Proof. Taking $a = b = 0$ gives $0 \in W$; special choices recover closure under addition and scalar multiplication. The converse is immediate. ☐

Theorem 2.2 (Dependence lemma)**.** *A list $v_1, \dots, v_r$ is dependent if and only if one vector lies in the span of the preceding vectors after a suitable reordering.*

Proof. Solve a nontrivial relation for a vector whose coefficient is nonzero. Conversely, any span relation rearranges to a nontrivial dependence relation. ☐

2.8 Worked examples

Example 2.3 (Solution spaces)**.** The solution set of a homogeneous linear system $Ax = 0$ is a subspace of $\mathbb{F}^n$ because matrix multiplication is linear.

Example 2.4 (Polynomial constraints)**.** Polynomials of degree at most 4 satisfying $p(1) = 0$ form a subspace: evaluation at 1 respects addition and scalar multiplication.

Example 2.5 (A nonexample)**.** The set $\{(x, y) : xy = 0\}$ is the union of the coordinate axes and is not a subspace because $(1, 0) + (0, 1) = (1, 1)$ leaves the set.

2.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;

2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

2.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 2.6 (Intersection and sum of planes)**.** Let $U = \{(x, y, z) : z = 0\}$ and $W = \{(x, y, z) : y = 0\}$. Find bases for $U \cap W$ and $U + W$.

Solution. The intersection requires $y = z = 0$, so $U \cap W = \operatorname{span}\{(1, 0, 0)\}$. Since $(1, 0, 0), (0, 1, 0) \in U$ and $(0, 0, 1) \in W$, the sum contains the standard basis and hence equals $\mathbb{R}^3$. □

Problem 2.7 (A union that fails)**.** Give an explicit failure of closure showing that the union of the x- and y-axes in $\mathbb{R}^2$ is not a subspace.

Solution. Both $(1, 0)$ and $(0, 1)$ lie in the union, but their sum $(1, 1)$ lies on neither axis. Therefore the union is not closed under addition. □

Problem 2.8 (Span from row reduction)**.** Determine whether $(2, 1, 3)$ lies in the span of $(1, 0, 1)$ and $(0, 1, 1)$.

Solution. Solve $a(1, 0, 1) + b(0, 1, 1) = (2, 1, 3)$. The first two coordinates force $a = 2$, $b = 1$, and then the third gives $a + b = 3$, which holds. Hence the vector lies in the span. □

Problem 2.9 (Redundant generator)**.** Show that $\operatorname{span}(u, v, u+2v) = \operatorname{span}(u, v)$ for arbitrary vectors u, v.

Solution. The third vector already lies in $\operatorname{span}(u, v)$, so adding it cannot enlarge the span. Conversely u and v occur in the larger generating list, so the smaller span is contained in the larger one. □

Problem 2.10 (Subspace defined by a functional)**.** Let $W = \{p \in \mathcal{P}_3 : p(1) = 0\}$. Prove W is a subspace and find a convenient spanning set.

Solution. Evaluation at 1 is linear, so W is its kernel and hence a subspace. Every p with $p(1) = 0$ is divisible by $x-1$, so $W = (x-1)\mathcal{P}_2$ and is spanned by $(x-1), x(x-1), x^2(x-1)$. □

Problem 2.11 (Dependence with a parameter)**.** For which a are $(1, a)$ and $(a, 1)$ linearly dependent in $\mathbb{R}^2$?

Solution. Two planar vectors are dependent exactly when the determinant $\begin{vmatrix} 1 & a \\ a & 1 \end{vmatrix} = 1 - a^2$ vanishes. Thus $a = \pm 1$. □

Problem 2.12 (Minimal spanning extraction)**.** From the list $(1, 0, 0), (0, 1, 0), (1, 1, 0), (0, 0, 1)$ extract a basis of its span.

Solution. The third vector is the sum of the first two, so it is redundant. The remaining vectors are the standard basis of $\mathbb{R}^3$, hence a basis of the span. □

Problem 2.13 (A nonlinear constraint)**.** Why is $\{(x, y) : xy = 0\}$ not a subspace even though it contains 0 and is closed under scalar multiplication?

Solution. It is not closed under addition: $(1,0)$ and $(0,1)$ satisfy the constraint, but their sum $(1,1)$ does not. All subspace conditions are necessary. ☐

Problem 2.14 (Independence of monomials)**.** Prove $1, x, \dots, x^n$ are linearly independent in $\mathcal{P}_n$ without invoking dimension.

Solution. If $\sum_{k=0}^n a_k x^k$ is the zero polynomial, then every coefficient of that polynomial is zero. Hence all $a_k = 0$, which is exactly linear independence. ☐

Problem 2.15 (Sum not direct)**.** Let $U = \operatorname{span}(e_1, e_2)$ and $W = \operatorname{span}(e_2, e_3)$ in $\mathbb{R}^3$. Explain why $U + W = \mathbb{R}^3$ but the sum is not direct.

Solution. The sum contains e_1, e_2, e_3, so it is all of $\mathbb{R}^3$. But e_2 is a nonzero vector in both U and W, so $U \cap W \neq \{0\}$; therefore the representation $u + w$ is not unique. ☐

Problem 2.16 (Solving a dependence relation)**.** Find a nontrivial relation among $(1,1,0), (0,1,1), (1,0,-1)$.

Solution. Adding the second and third gives $(1,1,0)$, so $-(1,1,0) + (0,1,1) + (1,0,-1) = 0$. Thus the list is dependent. ☐

Problem 2.17 (Subspace synthesis)**.** Prove that the solution set of any homogeneous system $Ax = 0$ is a subspace, and explain why the analogous set $Ax = b$ with $b \neq 0$ generally is not.

Solution. If $Ax = Ay = 0$, then $A(\alpha x + \beta y) = \alpha Ax + \beta Ay = 0$, so the homogeneous solution set is a subspace. If $Ax = b$ with $b \neq 0$, then $x = 0$ is not a solution, so the solution set cannot be a vector subspace; it is typically an affine translate of the kernel. ☐

2.11 Exercises with complete solutions

Exercise 2.18. Show that the intersection of any family of subspaces is a subspace.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The zero vector lies in every member. If two vectors lie in the intersection, every linear combination lies in each member and hence in the intersection. ☐

Exercise 2.19. Show that $U \cup W$ is a subspace only if $U \subseteq W$ or $W \subseteq U$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. If neither contains the other, choose $u \in U \setminus W$ and $w \in W \setminus U$. Then $u + w$ belongs to neither subspace, contradicting closure. ☐

Exercise 2.20. Find the span of $(1,0,1)$ and $(0,1,1)$ in $\mathbb{R}^3$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. It is $\{(a, b, a+b) : a, b \in \mathbb{R}\}$, equivalently the plane $z = x + y$. ☐

Exercise 2.21. Are $1 + x$, $x + x^2$, and $1 - x^2$ independent?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. No. The first minus the second minus the third is zero: $(1+x)-(x+x^2)-(1-x^2)=0$. $\square$

Exercise 2.22. Prove that a list containing the zero vector is dependent.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. Give the zero vector coefficient 1 and all other vectors coefficient 0. $\square$

Exercise 2.23. Determine whether $(1, 2)$ and $(2, 4)$ are independent.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. They are dependent because $(2, 4) = 2(1, 2)$. $\square$

Exercise 2.24. Show that the span of a set equals the intersection of all subspaces containing it.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. The span is itself a subspace containing the set, and every subspace containing the set contains all its finite linear combinations. $\square$

Exercise 2.25. Relate pivots in a matrix to independence of its columns.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. A homogeneous system $Ax = 0$ has only the zero solution exactly when every column variable is a pivot variable; this is equivalent to column independence. $\square$

Chapter summary

Subspaces are the linear subsets of a vector space, span is the smallest subspace generated by specified vectors, and linear independence is the condition that a generating description has no redundancy. These three ideas form the basic grammar of finite-dimensional reasoning. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 3

Bases and Dimension

A basis is simultaneously a generating system and a coordinate system with no redundancy. Finite-dimensional vector spaces become computationally manageable because every vector then has a unique finite coordinate list. Dimension records the size of any basis and is independent of the particular basis chosen.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use bases in basis-free and coordinate form;
- explain and use coordinates relative to a basis in basis-free and coordinate form;
- explain and use extension and reduction in basis-free and coordinate form;
- explain and use dimension in basis-free and coordinate form;
- explain and use dimension of subspaces in basis-free and coordinate form;
- explain and use dimension formula in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

3.1 Bases

A basis of V is a linearly independent list that spans V. The two conditions guarantee existence and uniqueness of coordinates.

3.2 Coordinates relative to a basis

If $B = (v_1, \dots, v_n)$ is a basis, every v has a unique representation $v = \sum a_i v_i$. The coordinate column is written $[v]_B$.

3.3 Extension and reduction

An independent list in a finite-dimensional space can be extended to a basis, while a spanning list can be reduced to a basis by deleting redundant vectors.

3.4 Dimension

All bases of a finite-dimensional vector space have the same number of vectors. That common number is $\dim V$.

3.5 Dimension of subspaces

If $W \subseteq V$ and V is finite-dimensional, then $\dim W \leq \dim V$, with equality only when $W = V$.

3.6 Dimension formula

For finite-dimensional subspaces $U, W \subseteq V$, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

3.7 Core structural results

Theorem 3.1 (Basis uniqueness of coordinates)**.** *A list is a basis if and only if every vector has a unique representation as a linear combination of the list.*

Proof. Spanning gives existence. If two representations exist, subtract them to obtain a dependence relation; independence forces equal coefficients. □

Theorem 3.2 (Dimension formula for sums)**.** *For finite-dimensional U, W, one has $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.*

Proof. Choose a basis of $U \cap W$, extend it separately to bases of U and W, and check that the union of the extensions spans $U + W$ and is independent. □

3.8 Worked examples

Example 3.3 (Polynomial dimension)**.** The space $\mathcal{P}_n$ of polynomials of degree at most n has basis $1, x, \ldots, x^n$ and dimension $n + 1$.

Example 3.4 (Symmetric matrices)**.** The real symmetric $n \times n$ matrices have dimension $n(n + 1)/2$, one free entry for each diagonal or upper-triangular position.

Example 3.5 (Trace-zero matrices)**.** The trace-zero subspace of $M_n(\mathbb{F})$ has dimension $n^2 - 1$.

3.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

3.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 3.6 (Basis test by uniqueness)**.** Show that $B = ((1,1),(1,-1))$ is a basis of $\mathbb{R}^2$ and find the coordinates of (a,b).

Solution. Solve $c_1(1,1) + c_2(1,-1) = (a,b)$. Then $c_1 + c_2 = a$ and $c_1 - c_2 = b$, giving $c_1 = (a+b)/2$, $c_2 = (a-b)/2$. Every vector has a unique representation, so B is a basis. □

Problem 3.7 (Dimension of a constrained polynomial space)**.** Find the dimension of $W = \{p \in \mathcal{P}_4 : p(0) = p(1) = 0\}$.

Solution. The conditions imply $p(x) = x(x-1)q(x)$ with $q \in \mathcal{P}_2$. Multiplication by $x(x-1)$ is injective, so W is isomorphic to $\mathcal{P}_2$ and has dimension 3. □

Problem 3.8 (Extend an independent list)**.** Extend $(1,1,0)$ and $(0,1,1)$ to a basis of $\mathbb{R}^3$.

Solution. The two vectors are independent. Choose, for instance, $e_1 = (1,0,0)$. It is not in their span because solving $a(1,1,0) + b(0,1,1) = e_1$ forces $a = 1$, $b = 0$, but then the second coordinate is 1. Thus the three vectors form a basis. □

Problem 3.9 (Reduce a spanning list)**.** Reduce $(1,0,0),(0,1,0),(1,1,0),(0,0,1)$ to a basis without changing its span.

Solution. Delete $(1,1,0)$ because it is the sum of the first two. The remaining three standard vectors are independent and span $\mathbb{R}^3$. □

Problem 3.10 (Dimension of symmetric matrices)**.** Compute $\dim \mathrm{Sym}_n(\mathbb{R})$.

Solution. A symmetric matrix is determined by its n diagonal entries and its $n(n-1)/2$ entries above the diagonal. Thus the dimension is $n + n(n-1)/2 = n(n+1)/2$. □

Problem 3.11 (Dimension formula in action)**.** If $\dim U = 5$, $\dim W = 4$, and $\dim(U+W) = 7$, find $\dim(U \cap W)$.

Solution. The dimension formula gives $7 = 5 + 4 - \dim(U \cap W)$, so the intersection has dimension 2. □

Problem 3.12 (Full-dimension subspace)**.** Prove that a subspace W of a finite-dimensional space V with $\dim W = \dim V$ must equal V.

Solution. A basis of W is an independent list in V with exactly $\dim V$ vectors. Any such list is a basis of V, so it spans V. Since it lies in W, we obtain $V \subseteq W$, hence equality. □

Problem 3.13 (Trace-zero matrices)**.** Why does the space of trace-zero $n \times n$ matrices have dimension $n^2 - 1$?

Solution. The trace map $M_n(\mathbb{F}) \to \mathbb{F}$ is a nonzero linear functional and hence surjective. Its kernel is the trace-zero subspace. Rank-nullity gives dimension $n^2 - 1$. □

Problem 3.14 (Coordinate uniqueness)**.** Suppose $B = (v_1, \ldots, v_n)$ is a basis and $\sum a_i v_i = \sum b_i v_i$. Prove $a_i = b_i$ for every i.

Solution. Subtract the two representations to obtain $\sum (a_i - b_i) v_i = 0$. Basis vectors are independent, so every coefficient $a_i - b_i$ is zero. □

Problem 3.15 (Finite versus infinite dimension)**.** Explain why the extension theorem for a finite independent list does not by itself prove every vector space has a basis.

Solution. In finite-dimensional spaces the extension process terminates because there is a finite spanning list. For arbitrary vector spaces, one needs a maximality principle such as Zorn's lemma to guarantee a maximal independent set; the elementary finite algorithm is insufficient. ☐

Problem 3.16 (Direct-sum dimension)**.** If $V = U \oplus W$ with finite dimensions, prove $\dim V = \dim U + \dim W$.

Solution. Take bases of U and W. Their union spans V because every vector decomposes as $u + w$. If a linear combination of the union is zero, the U-part equals minus the W-part and hence lies in $U \cap W = \{0\}$, forcing all coefficients to vanish. ☐

Problem 3.17 (Basis synthesis)**.** A list of n vectors in an n-dimensional space spans. Why is it automatically a basis?

Solution. If the list were dependent, one vector could be removed without changing the span, producing a spanning list of fewer than n vectors, impossible in an n-dimensional space. Hence the list is independent and therefore a basis. ☐

3.11 Exercises with complete solutions

Exercise 3.18. Show that $(1, 0), (1, 1)$ is a basis of $\mathbb{R}^2$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The two vectors are not scalar multiples, so they are independent; two independent vectors in a two-dimensional space form a basis. ☐

Exercise 3.19. Find the coordinates of $(3, 2)$ in the basis $((1, 0), (1, 1))$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Solve $a(1, 0) + b(1, 1) = (3, 2)$. Then $b = 2$ and $a = 1$, so the coordinate column is $(1, 2)^T$. ☐

Exercise 3.20. Find a basis for polynomials in $\mathcal{P}_3$ satisfying $p(0) = 0$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The condition kills the constant term, so x, x^2, x^3 is a basis. ☐

Exercise 3.21. Why can no four vectors in $\mathbb{R}^3$ be independent?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Any independent list has length at most the dimension, which is 3. ☐

Exercise 3.22. If $U \subseteq V$ and $\dim U = \dim V < \infty$, prove $U = V$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. A basis of U is an independent list in V of full dimension, hence already a basis of V. ☐

Exercise 3.23. Compute the dimension of skew-symmetric $n \times n$ matrices over a field of characteristic not 2.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Diagonal entries are zero and entries below the diagonal are determined by those above it, giving $n(n-1)/2$. ☐

Exercise 3.24. Suppose $\dim U = 3$, $\dim W = 4$, and $\dim(U \cap W) = 2$. Find $\dim(U + W)$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The dimension formula gives $3 + 4 - 2 = 5$. ☐

Exercise 3.25. Explain why every finite spanning set contains a basis.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Repeatedly delete any vector lying in the span of the others. The process terminates and preserves span; the remaining list is independent. ☐

Chapter summary

A basis is simultaneously a generating system and a coordinate system with no redundancy. Finite-dimensional vector spaces become computationally manageable because every vector then has a unique finite coordinate list. Dimension records the size of any basis and is independent of the particular basis chosen. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 4

Coordinates and Change of Basis

Coordinates turn an abstract vector space into a copy of $\mathbb{F}^n$, but the copy depends on a chosen basis. Change-of-basis matrices explain precisely how two coordinate descriptions of the same vector are related. This is the first systematic appearance of conjugation and covariance.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use coordinate isomorphisms in basis-free and coordinate form;
- explain and use transition matrices in basis-free and coordinate form;
- explain and use inverse transitions in basis-free and coordinate form;
- explain and use coordinate covariants in basis-free and coordinate form;
- explain and use dual coordinates in basis-free and coordinate form;
- explain and use conditioning of bases in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

4.1 Coordinate isomorphisms

For an ordered basis $B = (v_1, \ldots, v_n)$, the coordinate map $\kappa_B : V \to \mathbb{F}^n$, $v \mapsto [v]_B$, is a linear isomorphism.

4.2 Transition matrices

Given bases B, C, the matrix $P_{C\leftarrow B}$ is defined by $[v]_C = P_{C\leftarrow B}[v]_B$. Its columns are the C-coordinates of the vectors of B.

4.3 Inverse transitions

Changing from B to C and back gives $P_{B\leftarrow C} = P_{C\leftarrow B}^{-1}$.

4.4 Coordinate covariants

Vectors and linear maps do not change when the basis changes; their coordinate arrays do. Good notation separates the object from its representation.

4.5 Dual coordinates

Linear functionals transform by the inverse transpose of the vector change-of-basis matrix. This anticipates the distinction between vectors and covectors.

4.6 Conditioning of bases

A mathematically valid basis can be numerically poor if its vectors are nearly dependent. Orthogonal bases mitigate this instability.

4.7 Core structural results

Theorem 4.1 (Transition matrix rule). *The columns of $P_{C\leftarrow B}$ are $[v_1]_C, \ldots, [v_n]_C$ when $B = (v_1, \ldots, v_n)$.*

Proof. Apply the transition identity to each basis vector. Its B-coordinate is the corresponding standard vector, so the resulting C-coordinate is the corresponding column. □

Theorem 4.2 (Cocycle identity). *For bases B, C, D, $P_{D\leftarrow B} = P_{D\leftarrow C} P_{C\leftarrow B}$.*

Proof. Apply the two coordinate changes successively to an arbitrary coordinate vector and use uniqueness of the representing matrix. □

4.8 Worked examples

Example 4.3 (Two planar bases). For $B = ((1, 0), (1, 1))$ and standard E, $P_{E\leftarrow B} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Example 4.4 (Polynomial bases). The monomial basis and shifted basis $1, (x - 1), (x - 1)^2$ of $\mathcal{P}_2$ are related by expanding powers of $x - 1$.

Example 4.5 (Scaling a basis). Replacing v_i by $c_i v_i$ multiplies the corresponding transition column by c_i.

4.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;

4. know which hypotheses are essential and produce a counterexample when one is removed.

4.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 4.6 (Transition matrix from columns)**.** For $B = ((1,1),(1,-1))$ and the standard basis E, compute $P_{E\leftarrow B}$ and its inverse.

Solution. The columns are the E-coordinates of the B-vectors, so $P_{E\leftarrow B} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Its inverse is $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$, which is $P_{B\leftarrow E}$. □

Problem 4.7 (Coordinates of a polynomial)**.** Find the coordinates of $x^2 + 1$ in $B = (1, x-1, (x-1)^2)$.

Solution. Write $x = (x-1)+1$. Then $x^2+1 = (x-1)^2 + 2(x-1) + 2$, so the coordinate vector is $(2,2,1)^T$. □

Problem 4.8 (Cocycle identity)**.** Given bases B, C, D, prove $P_{D\leftarrow B} = P_{D\leftarrow C}P_{C\leftarrow B}$.

Solution. For every vector v, first change B-coordinates to C-coordinates and then to D-coordinates. Thus $[v]_D = P_{D\leftarrow C}P_{C\leftarrow B}[v]_B$. Uniqueness of the matrix realizing this coordinate transformation gives the identity. □

Problem 4.9 (A badly conditioned basis)**.** Why is $B_\varepsilon = ((1,0),(1,\varepsilon))$ a basis for $\varepsilon \neq 0$ but numerically poor when $|\varepsilon|$ is small?

Solution. The determinant of the basis matrix is ε, so it is invertible for $\varepsilon \neq 0$. Its inverse contains entries of size $1/\varepsilon$, so small perturbations in coordinates can be greatly amplified as $\varepsilon \to 0$. □

Problem 4.10 (Permutation of basis vectors)**.** If $C = (v_2, v_1, v_3)$ is obtained from $B = (v_1, v_2, v_3)$ by swapping the first two vectors, describe $P_{C\leftarrow B}$.

Solution. It is the permutation matrix that swaps the first two coordinates: $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$. The same matrix is its inverse. □

Problem 4.11 (Scaling a basis)**.** Let $C = (2v_1, 3v_2)$ when $B = (v_1, v_2)$. Find $P_{C\leftarrow B}$.

Solution. If $v = av_1 + bv_2 = c(2v_1) + d(3v_2)$, then $c = a/2$ and $d = b/3$. Thus $P_{C\leftarrow B} = \operatorname{diag}(1/2, 1/3)$. □

Problem 4.12 (Coordinates versus objects)**.** Two coordinate columns for the same vector are different. Explain why this does not contradict uniqueness of coordinates.

Solution. Coordinates are unique only after the basis is fixed. Different bases define different coordinate isomorphisms $V \to \mathbb{F}^n$, so the same abstract vector can have different coordinate columns without ambiguity. □

Problem 4.13 (Dual transformation)**.** Suppose vector coordinates change by $x_C = Px_B$. How must a row covector ℓ change so that the scalar $\ell(x)$ is invariant?

Solution. We require $\ell_C x_C = \ell_B x_B$ for all x_B. Since $x_C = P x_B$, this gives $\ell_C P = \ell_B$, hence $\ell_C = \ell_B P^{-1}$. In column-covector convention this becomes the inverse transpose. $\square$

Problem 4.14 (Coordinate map isomorphism)**.** Prove the coordinate map $\kappa_B : V \to \mathbb{F}^n$ is linear and bijective.

Solution. Linearity follows by combining coefficients in the basis expansion. Every coordinate vector gives a linear combination of the basis vectors, so the map is onto. Uniqueness of basis coordinates gives injectivity. $\square$

Problem 4.15 (Three-basis computation)**.** If $P_{C\leftarrow B} = A$ and $P_{D\leftarrow C} = B$, what is $P_{B\leftarrow D}$?

Solution. First $P_{D\leftarrow B} = BA$. Therefore the reverse transition is $(BA)^{-1} = A^{-1}B^{-1}$. $\square$

Problem 4.16 (Basis from columns)**.** Why does every invertible matrix P determine a new ordered basis relative to a fixed basis E?

Solution. Take as new basis vectors the vectors whose E-coordinate columns are the columns of P. Invertibility means those columns are independent and span the whole space, so they form a basis. $\square$

Problem 4.17 (Coordinate synthesis)**.** Explain why changing basis is not the same operation as applying a linear transformation to vectors.

Solution. A basis change changes only the description of the same vector: the geometric object is fixed and coordinates are transformed by a transition matrix. Applying an operator changes the vector itself. Confusing these roles is the source of many incorrect conjugation formulas. $\square$

4.11 Exercises with complete solutions

Exercise 4.18. Compute $P_{E\leftarrow B}$ for $B = ((1,1),(1,-1))$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. Its columns are the basis vectors in standard coordinates: $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. $\square$

Exercise 4.19. Find $P_{B\leftarrow E}$ for the preceding basis.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. Invert the matrix to get $\frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. $\square$

Exercise 4.20. Verify the cocycle identity for three arbitrary bases.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. $\square$

Solution. For every v, substitute $[v]_C = P_{C\leftarrow B}[v]_B$ into $[v]_D = P_{D\leftarrow C}[v]_C$; equality for all coordinate vectors gives the matrix identity. $\square$

Exercise 4.21. Why must every transition matrix be invertible?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. It represents the composition of two coordinate isomorphisms, hence an isomorphism. □

Exercise 4.22. Compute coordinates of $p(x) = x^2$ in $B = (1, x-1, (x-1)^2)$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Since $x = (x-1)+1$, $x^2 = (x-1)^2+2(x-1)+1$, so the coordinates are $(1,2,1)^T$. □

Exercise 4.23. How do coordinates change if the basis order is permuted?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The coordinate column is permuted inversely to the basis order; the transition is represented by a permutation matrix. □

Exercise 4.24. Explain why the vector itself is basis-independent.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. A basis selects a coordinate isomorphism but does not alter the element $v \in V$; only its numerical representative changes. □

Exercise 4.25. Why are nearly parallel planar basis vectors numerically problematic?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The transition matrix is close to singular, so its inverse can have large entries and amplify small data errors. □

Chapter summary

Coordinates turn an abstract vector space into a copy of $\mathbb{F}^n$, but the copy depends on a chosen basis. Change-of-basis matrices explain precisely how two coordinate descriptions of the same vector are related. This is the first systematic appearance of conjugation and covariance. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 5

Linear Transformations

Linear transformations are the morphisms of vector spaces: they preserve exactly the operations that define linear structure. Once linearity is recognized, a map is determined by its action on a basis, compositions remain linear, and apparently different formulas become instances of the same operator calculus.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use definition and first consequences in basis-free and coordinate form;
- explain and use determination by a basis in basis-free and coordinate form;
- explain and use algebra of operators in basis-free and coordinate form;
- explain and use composition in basis-free and coordinate form;
- explain and use inverse maps in basis-free and coordinate form;
- explain and use important operators in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

5.1 Definition and first consequences

A map $T : V \to W$ is linear if $T(av + bw) = aT(v) + bT(w)$. Then $T(0) = 0$ and all finite linear combinations pass through T.

5.2 Determination by a basis

A linear map is uniquely determined by its values on a basis of the domain. Conversely, arbitrary images of basis vectors extend uniquely to a linear map.

5.3 Algebra of operators

For fixed V, W, the set $\mathcal{L}(V, W)$ is a vector space under pointwise addition and scalar multiplication. Endomorphisms $\mathcal{L}(V)$ also compose.

5.4 Composition

The composite of linear maps is linear. Composition is associative and distributes over addition but is generally not commutative.

5.5 Inverse maps

A bijective linear map has a linear inverse. Such maps are isomorphisms and identify vector spaces as having the same linear structure.

5.6 Important operators

Differentiation on polynomials, evaluation functionals, coordinate projections, inclusion maps, shifts, and matrix multiplication are canonical examples.

5.7 Core structural results

Theorem 5.1 (Extension from a basis). *If $B = (v_1, \ldots, v_n)$ is a basis of V and $w_i \in W$ are arbitrary, there is a unique linear $T : V \to W$ with $T(v_i) = w_i$.*

Proof. Write each v uniquely as $\sum a_i v_i$ and define $T(v) = \sum a_i w_i$. Uniqueness follows because any linear map must use this formula. □

Theorem 5.2 (Inverse of an isomorphism). *If $T : V \to W$ is linear and bijective, then $T^{-1} : W \to V$ is linear.*

Proof. Write $y_j = T(x_j)$. Then $T^{-1}(ay_1 + by_2) = T^{-1}(T(ax_1 + bx_2)) = ax_1 + bx_2$. □

5.8 Worked examples

Example 5.3 (Differentiation). $D : \mathcal{P}_n \to \mathcal{P}_{n-1}$, $D(p) = p'$, is linear because differentiation respects sums and scalar multiples.

Example 5.4 (Evaluation). For fixed a, $E_a(p) = p(a)$ is a linear functional on a polynomial space.

Example 5.5 (Shear). $T(x, y) = (x + y, y)$ is linear and invertible, with inverse $(u, v) \mapsto (u - v, v)$.

5.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;

3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

5.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 5.6 (Basis images determine a map)**.** Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ satisfy $T(e_1) = (1, 0, 1)$ and $T(e_2) = (0, 1, 1)$. Compute $T(x, y)$.

Solution. Linearity gives $T(x, y) = xT(e_1) + yT(e_2) = (x, y, x + y)$. No other linear map can have the prescribed basis images. □

Problem 5.7 (Detecting nonlinearity quickly)**.** Is $F(x, y) = (x + y, xy)$ linear?

Solution. No. The second component is quadratic. For instance $F(2, 0) = (2, 0)$ while $2F(1, 0) = (2, 0)$ happens to agree, but $F(1, 1) = (2, 1)$ whereas $F(1, 0) + F(0, 1) = (2, 0)$, so additivity fails. □

Problem 5.8 (Differentiation as an operator)**.** On $\mathcal{P}_3$, determine $\ker D$ and $\operatorname{im} D$ for $D(p) = p'$.

Solution. The derivative is zero exactly for constant polynomials, so $\ker D = \mathcal{P}_0$. Every polynomial of degree at most 2 has an antiderivative in $\mathcal{P}_3$, so $\operatorname{im} D = \mathcal{P}_2$. □

Problem 5.9 (Composition order)**.** Let $P(x, y) = (x, 0)$ and $R(x, y) = (-y, x)$. Compare RP and PR.

Solution. $RP(x, y) = R(x, 0) = (0, x)$, whereas $PR(x, y) = P(-y, x) = (-y, 0)$. Thus composition is not commutative. □

Problem 5.10 (A nilpotent map)**.** Define $N(x, y, z) = (y, z, 0)$. Determine the smallest k with $N^k = 0$.

Solution. $N^2(x, y, z) = (z, 0, 0)$ and $N^3 = 0$, while $N^2 \neq 0$. Thus the nilpotency index is 3. □

Problem 5.11 (Evaluation functional)**.** Show $E_1 : \mathcal{P}_2 \to \mathbb{R}$, $E_1(p) = p(1)$, is linear and find its values on $1, x, x^2$.

Solution. Evaluation respects sums and scalar multiples. Each of $1, x, x^2$ takes value 1 at $x = 1$, so the functional is determined by the coordinate row $(1, 1, 1)$. □

Problem 5.12 (Inverse linearity)**.** If T is a bijective linear map, prove T^{-1} is linear.

Solution. Write $y_i = T(x_i)$. Then $T^{-1}(ay_1 + by_2) = T^{-1}(T(ax_1 + bx_2)) = ax_1 + bx_2 = aT^{-1}y_1 + bT^{-1}y_2$. □

Problem 5.13 (Zero map versus zero vector)**.** Explain the difference between the zero vector in W and the zero element of $\mathcal{L}(V, W)$.

Solution. The zero vector is a single element 0_W. The zero operator is a function $0 : V \to W$ sending every v to 0_W. It is the additive identity in the vector space of linear maps. □

Problem 5.14 (A projection decomposition)**.** Let $P(x, y, z) = (x, y, 0)$. Show $P^2 = P$ and describe its fixed vectors and kernel.

Solution. Applying P twice changes nothing after the first application, so $P^2 = P$. Fixed vectors satisfy $z = 0$, the xy-plane; the kernel is the z-axis. ☐

Problem 5.15 (A linear map from a formula)**.** For $T(A) = A + A^T$ on $M_2(\mathbb{R})$, prove linearity and characterize the image.

Solution. Transpose is linear, so $T(\alpha A + \beta B) = \alpha T(A) + \beta T(B)$. Every output is symmetric. Conversely, if S is symmetric then $T(S/2) = S$, so the image is exactly the symmetric matrices. ☐

Problem 5.16 (Restriction to a subspace)**.** Let $T(x, y, z) = (x + y, y + z, z)$. Is the plane $U = \{z = 0\}$ invariant?

Solution. If $(x, y, 0) \in U$, then $T(x, y, 0) = (x + y, y, 0) \in U$. Hence the restriction $T|_U$ is a well-defined linear operator on U. ☐

Problem 5.17 (Operator synthesis)**.** Why does specifying values on a spanning list fail to define a linear map unless all dependence relations are respected?

Solution. If $\sum a_i v_i = 0$ is a relation in the spanning list, linearity requires $\sum a_i T(v_i) = 0$. Arbitrary assigned images may violate this condition, giving contradictory values for the same vector. A basis avoids the problem because it has no nontrivial relations. ☐

5.11 Exercises with complete solutions

Exercise 5.18. Show that $T(x, y) = (x + 2y, 3x - y)$ is linear.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Each output coordinate is a homogeneous linear expression in x, y, so direct substitution verifies preservation of linear combinations. ☐

Exercise 5.19. Is $T(x, y) = (x + 1, y)$ linear?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. No. A linear map must send 0 to 0, but this map sends $(0, 0)$ to $(1, 0)$. ☐

Exercise 5.20. Construct the unique linear map with $T(e_1) = (1, 1)$ and $T(e_2) = (2, 0)$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. For $(x, y) = xe_1 + ye_2$, $T(x, y) = x(1, 1) + y(2, 0) = (x + 2y, x)$. ☐

Exercise 5.21. Show that composition of linear maps is linear.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Apply the inner map to a linear combination, then use linearity of the outer map. ☐

Exercise 5.22. Prove that two linear maps agreeing on a basis agree everywhere.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Every vector is a linear combination of basis vectors, and both maps distribute over that combination. ☐

Exercise 5.23. Give a nonzero nilpotent operator on $\mathbb{R}^2$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Take $N(x, y) = (y, 0)$. Then $N \neq 0$ but $N^2 = 0$. ☐

Exercise 5.24. Show that $\mathcal{L}(V, W)$ is a vector space.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Verify the vector-space axioms pointwise; the zero vector is the zero map. ☐

Exercise 5.25. Explain why a bijective nonlinear map need not be an isomorphism of vector spaces.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. An isomorphism must preserve addition and scalar multiplication; bijectivity alone preserves only set structure. ☐

Chapter summary

Linear transformations are the morphisms of vector spaces: they preserve exactly the operations that define linear structure. Once linearity is recognized, a map is determined by its action on a basis, compositions remain linear, and apparently different formulas become instances of the same operator calculus. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 6

Kernels, Images and Isomorphisms

Kernel and image measure exactly how a linear transformation fails to be injective or surjective. The rank–nullity theorem balances those two defects and turns structural questions into dimension counts. Quotient spaces then package the first isomorphism theorem in its natural form.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use kernel in basis-free and coordinate form;
- explain and use image and rank in basis-free and coordinate form;
- explain and use rank–nullity in basis-free and coordinate form;
- explain and use injective versus surjective in basis-free and coordinate form;
- explain and use quotients in basis-free and coordinate form;
- explain and use first isomorphism theorem in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

6.1 Kernel

$\ker T = \{v : T(v) = 0\}$ is a subspace of the domain. A linear map is injective exactly when its kernel is zero.

6.2 Image and rank

$\operatorname{im} T = \{T(v) : v \in V\}$ is a subspace of the codomain. Its dimension is the rank.

6.3 Rank–nullity

For finite-dimensional V, $\dim V = \dim\ker T + \dim\operatorname{im} T$.

6.4 Injective versus surjective

For a linear endomorphism of a finite-dimensional space, injective, surjective and bijective are equivalent.

6.5 Quotients

For a subspace $U \subseteq V$, the quotient V/U consists of cosets $v + U$. It has dimension $\dim V - \dim U$ in finite dimensions.

6.6 First isomorphism theorem

The map $V/\ker T \to \operatorname{im} T$, $v + \ker T \mapsto T(v)$, is a well-defined linear isomorphism.

6.7 Core structural results

Theorem 6.1 (Rank–nullity theorem)**.** *If $T : V \to W$ and V is finite-dimensional, then $\dim V = \dim\ker T + \dim\operatorname{im} T$.*

Proof. Choose a basis of the kernel and extend it to a basis of V. The images of the added basis vectors form a basis of the image. □

Theorem 6.2 (First isomorphism theorem)**.** *Every linear map induces an isomorphism $V/\ker T \cong \operatorname{im} T$.*

Proof. The induced map is well-defined because changing a representative by a kernel vector does not change its image; injectivity and surjectivity are immediate. □

6.8 Worked examples

Example 6.3 (Differentiation)**.** On $\mathcal{P}_n$, differentiation has kernel the constants and image $\mathcal{P}_{n-1}$, so nullity 1 plus rank n equals $n + 1$.

Example 6.4 (Projection)**.** The projection $\mathbb{R}^3 \to \mathbb{R}^2$, $(x, y, z) \mapsto (x, y)$, has one-dimensional kernel and rank 2.

Example 6.5 (Trace functional)**.** $\operatorname{tr} : M_n \to \mathbb{F}$ is surjective; its kernel has dimension $n^2 - 1$.

6.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

6.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 6.6 (Rank-nullity from equations)**.** For $T : \mathbb{R}^4 \to \mathbb{R}^2$, $T(x_1, x_2, x_3, x_4) = (x_1 + x_3, x_2 + x_4)$, find rank and nullity.

Solution. The map is onto because $(a, b) = T(a, b, 0, 0)$, so rank is 2. The kernel is given by $x_3 = -x_1$, $x_4 = -x_2$, hence has dimension 2. Rank plus nullity is 4. ☐

Problem 6.7 (Kernel detects injectivity)**.** Prove that a linear map is injective iff its kernel is $\{0\}$.

Solution. If $T(v) = T(w)$, then $T(v - w) = 0$; trivial kernel forces $v = w$. Conversely, if T is injective and $T(v) = 0 = T(0)$, then $v = 0$. ☐

Problem 6.8 (First isomorphism theorem)**.** For $T(x, y, z) = (x, y)$, identify $\mathbb{R}^3/\ker T$ with $\mathbb{R}^2$.

Solution. The kernel is the z-axis. Map the coset $(x, y, z)+\ker T$ to (x, y). This is well-defined, linear, injective, and onto; vertical displacement is exactly what the quotient forgets. ☐

Problem 6.9 (Trace kernel)**.** Compute the dimension of $\{A \in M_3(\mathbb{R}) : \operatorname{tr} A = 0\}$.

Solution. Trace is a nonzero linear functional from a 9-dimensional space to $\mathbb{R}$, hence has rank 1. Rank-nullity gives kernel dimension 8. ☐

Problem 6.10 (Image of differentiation)**.** For $D : \mathcal{P}_4 \to \mathcal{P}_3$, determine rank and nullity.

Solution. The kernel consists of constants, dimension 1. Every cubic has an antiderivative of degree at most 4, so D is onto $\mathcal{P}_3$, which has dimension 4. Thus rank 4, nullity 1. ☐

Problem 6.11 (Injective endomorphism)**.** Why is an injective linear endomorphism of a finite-dimensional space automatically surjective?

Solution. If $\dim V = n$ and the kernel is zero, rank-nullity gives rank n. The image is an n-dimensional subspace of the n-dimensional codomain V, so it equals V. ☐

Problem 6.12 (A quotient dimension)**.** If $\dim V = 10$ and $\dim U = 4$, compute $\dim(V/U)$.

Solution. Choose a basis of U and extend it to a basis of V. The six cosets of the added basis vectors form a basis of V/U, so the quotient dimension is $10 - 4 = 6$. ☐

Problem 6.13 (Rank under composition)**.** Show $\operatorname{rank}(ST) \le \min(\operatorname{rank} S, \operatorname{rank} T)$.

Solution. The image of ST is $S(\operatorname{im} T)$, so its dimension cannot exceed $\dim \operatorname{im} T$. It is also a subspace of $\operatorname{im} S$, so it cannot exceed $\operatorname{rank} S$. ☐

Problem 6.14 (Same kernel, different maps)**.** Give two nonzero linear maps $\mathbb{R}^2 \to \mathbb{R}$ with the same kernel.

Solution. Take $T(x, y) = x$ and $S(x, y) = 2x$. Both have kernel the y-axis, but they are different maps. The kernel alone does not determine a linear map. ☐

Problem 6.15 (Surjective map splitting)**.** Let $T : V \to W$ be surjective with finite dimensions. Show there is a subspace $U \subset V$ such that $T|_U : U \to W$ is an isomorphism.

Solution. Choose a basis of $\ker T$ and extend it to a basis of V. Let U be the span of the added vectors. Rank-nullity shows their number is $\dim W$, and their images form a basis of W, so the restriction is bijective. ☐

Problem 6.16 (Kernel of a matrix operator)**.** Let $L_A : M_2(\mathbb{R}) \to M_2(\mathbb{R})$ be $L_A(X) = AX$ with invertible A. Find its kernel and rank.

Solution. If $AX = 0$, multiply by A^{-1} to get $X = 0$, so the kernel is trivial. The domain has dimension 4, hence rank is 4; in fact L_A is an isomorphism. ☐

Problem 6.17 (Structural synthesis)**.** Explain why rank-nullity is a quantitative form of the first isomorphism theorem.

Solution. The first isomorphism theorem says $V/\ker T \cong \operatorname{im} T$. Taking finite dimensions gives $\dim V - \dim\ker T = \dim\operatorname{im} T$, which rearranges exactly to rank-nullity. ☐

6.11 Exercises with complete solutions

Exercise 6.18. Find the kernel of $T(x, y, z) = (x + y, y + z)$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Solve $x + y = 0$ and $y + z = 0$: $(x, y, z) = t(-1, 1, -1)$, so the kernel is one-dimensional. ☐

Exercise 6.19. Find the rank of the preceding map.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The domain has dimension 3 and the kernel dimension is 1, so rank–nullity gives rank 2. ☐

Exercise 6.20. Prove that zero kernel is equivalent to injectivity.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. If $T(v) = T(w)$ then $T(v - w) = 0$. Zero kernel forces $v = w$; the converse follows by comparing $T(v) = T(0)$. ☐

Exercise 6.21. Why does surjectivity imply rank equals codomain dimension?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The image equals the codomain, so their dimensions are equal. ☐

Exercise 6.22. Show $\dim(V/U) = \dim V - \dim U$ in finite dimensions.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Extend a basis of U to a basis of V; the cosets of the added basis vectors form a basis of the quotient. ☐

Exercise 6.23. Compute the nullity of the trace map on $M_3(\mathbb{R})$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The domain has dimension 9, the trace map has rank 1, so nullity is 8. ☐

Exercise 6.24. For an endomorphism of an n-dimensional space, show injective implies surjective.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Injectivity gives nullity 0, so rank–nullity gives rank n, which equals the codomain dimension. ☐

Exercise 6.25. Explain geometrically what quotienting $\mathbb{R}^3$ by the z-axis does.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Vectors differing by a vertical displacement are identified; the quotient is two-dimensional and can be identified with the xy-plane. ☐

Chapter summary

Kernel and image measure exactly how a linear transformation fails to be injective or surjective. The rank–nullity theorem balances those two defects and turns structural questions into dimension counts. Quotient spaces then package the first isomorphism theorem in its natural form. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Part II

Matrices and Operators

Chapter 7

Matrix Representation of Linear Maps

Matrices are coordinate representations of linear maps. The matrix is not the transformation itself: it depends on chosen bases in the domain and codomain. Once this distinction is kept visible, matrix multiplication, change of basis, rank and similarity all acquire transparent conceptual meanings.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use matrix of a linear map in basis-free and coordinate form;
- explain and use column rule in basis-free and coordinate form;
- explain and use composition and multiplication in basis-free and coordinate form;
- explain and use identity and inverses in basis-free and coordinate form;
- explain and use change of basis for operators in basis-free and coordinate form;
- explain and use rank and row reduction in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

7.1 Matrix of a linear map

For bases B of V and C of W, the matrix $[T]_{C\leftarrow B}$ is defined by $[T(v)]_C = [T]_{C\leftarrow B}[v]_B$.

7.2 Column rule

The jth column of $[T]_{C\leftarrow B}$ is $[T(v_j)]_C$.

7.3 Composition and multiplication

Coordinates turn composition into matrix multiplication: $[S \circ T] = [S][T]$ with compatible bases.

7.4 Identity and inverses

The identity has the identity matrix in the same basis. An isomorphism is represented by an invertible matrix.

7.5 Change of basis for operators

For an endomorphism, changing from basis B to C gives $[T]_C = P_{C \leftarrow B}[T]_B P_{B \leftarrow C}$. Thus representations of the same operator are similar.

7.6 Rank and row reduction

The rank of a matrix equals the dimension of the image of its associated linear map. Row operations preserve rank.

7.7 Core structural results

Theorem 7.1 (Matrix composition rule)**.** *With compatible bases,* $[S \circ T] = [S][T]$.

Proof. Apply both sides to an arbitrary coordinate vector and use the defining representation identity twice. □

Theorem 7.2 (Similarity under basis change)**.** *If* $A = [T]_B$ *and* $P = P_{C \leftarrow B}$*, then* $[T]_C = PAP^{-1}$.

Proof. Use $[v]_C = P[v]_B$ and express the coordinate identity for $T(v)$ in the two bases. □

7.8 Worked examples

Example 7.3 (Derivative matrix)**.** On $\mathcal{P}_2$ in the monomial basis, differentiation is represented by $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}$.

Example 7.4 (Rotation)**.** A planar rotation through angle θ has standard matrix $\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$.

Example 7.5 (Projection)**.** Projection onto the first coordinate axis has matrix $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$.

7.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;

2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

7.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 7.6 (Matrix from basis images)**.** Find the standard matrix of $T(x, y, z) = (x + z, 2y, x - z)$.

Solution. The columns are $T(e_1) = (1, 0, 1)$, $T(e_2) = (0, 2, 0)$, and $T(e_3) = (1, 0, -1)$, giving $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & -1 \end{pmatrix}$. ☐

Problem 7.7 (Composition matrix)**.** Let $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. Which matrix represents first applying A and then B?

Solution. Column vectors are acted on from the left, so first A then B is represented by BA. Matrix multiplication encodes composition in reverse verbal order. ☐

Problem 7.8 (Similarity from basis change)**.** If $[T]_B = A$ and $P = P_{C \leftarrow B}$, derive $[T]_C = PAP^{-1}$.

Solution. For $x_C = Px_B$, we have $[T(v)]_C = PAx_B = PAP^{-1}x_C$. Since this holds for every coordinate column, the C-matrix is PAP^{-1}. ☐

Problem 7.9 (Derivative matrix)**.** Write the matrix of $D : \mathcal{P}_3 \to \mathcal{P}_2$ in monomial bases.

Solution. $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, $D(x^3) = 3x^2$. Thus the columns are $(0, 0, 0)^T, (1, 0, 0)^T, (0, 2, 0)^T, (0, 0, 3)^T$. ☐

Problem 7.10 (Rank from columns)**.** For $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$ find the rank.

Solution. Every column is a scalar multiple of $(1, 2)^T$, and the matrix is nonzero. Hence the column space is one-dimensional and rank is 1. ☐

Problem 7.11 (Elementary matrix meaning)**.** What linear map does left multiplication by the elementary matrix that swaps rows 1 and 2 represent?

Solution. It composes a matrix-valued coordinate representation with the coordinate permutation swapping the first two output coordinates. Algebraically it performs the corresponding row operation. ☐

Problem 7.12 (Inverse matrix as inverse map)**.** Why does A^{-1} represent T^{-1} when A represents an isomorphism T in fixed bases?

Solution. The composition $T^{-1}T = I$ gives $[T^{-1}]A = I$ by the composition rule, so $[T^{-1}] = A^{-1}$. ☐

Problem 7.13 (Rectangular matrix)**.** What does a 3×2 matrix represent geometrically?

Solution. It represents a linear map from a two-dimensional coordinate space to a three-dimensional coordinate space. Its two columns are the images of the two domain basis vectors expressed in the codomain basis. □

Problem 7.14 (Same operator, different matrices)**.** Can two different matrices represent the same linear operator?

Solution. Yes, if different bases are used. The matrices are related by similarity for an endomorphism. With the basis fixed, however, the representing matrix is unique. □

Problem 7.15 (Projection matrix)**.** Find the standard matrix for projection onto $\mathrm{span}(e_1, e_3)$ in $\mathbb{R}^3$.

Solution. The map sends (x, y, z) to $(x, 0, z)$, so the matrix is $\mathrm{diag}(1, 0, 1)$. □

Problem 7.16 (Column space and image)**.** Explain why the column space of a matrix is the coordinate image of its linear map.

Solution. Every output is $Ax = \sum_j x_j a_j$, a linear combination of the columns a_j. Conversely every such combination occurs for the coefficient vector x, so the image equals the column span. □

Problem 7.17 (Matrix synthesis)**.** Why is similarity an equivalence relation on square matrices?

Solution. Reflexivity uses I, symmetry uses P^{-1}, and transitivity follows because $B = PAP^{-1}$ and $C = QBQ^{-1}$ imply $C = (QP)A(QP)^{-1}$. It therefore captures 'same operator in different bases'. □

7.11 Exercises with complete solutions

Exercise 7.18. Find the matrix of $T(x, y) = (x + 2y, 3x - y)$ in the standard basis.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The columns are $T(e_1) = (1, 3)$ and $T(e_2) = (2, -1)$, so $\begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$. □

Exercise 7.19. Why do columns record images of basis vectors?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The coordinate vector of the jth basis vector is e_j, and multiplying by a matrix selects its jth column. □

Exercise 7.20. Prove that the matrix of an inverse map is the inverse matrix.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. From $T^{-1}T = I$, the composition rule gives $[T^{-1}][T] = I$. □

Exercise 7.21. Compute the rank of $\begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The second row and second column are scalar multiples of the first, while the matrix is nonzero, so rank is 1. ☐

Exercise 7.22. Show that similar matrices have the same rank.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Left and right multiplication by invertible matrices correspond to isomorphisms and do not change image dimension. ☐

Exercise 7.23. Write the standard matrix of reflection across the x-axis.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$. ☐

Exercise 7.24. Explain why matrix multiplication is generally noncommutative.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Composition of transformations is order-sensitive; for example, a projection followed by a rotation differs from the reverse order. ☐

Exercise 7.25. Relate elementary row operations to invertible matrices.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Each elementary row operation is left multiplication by an elementary invertible matrix. ☐

Chapter summary

Matrices are coordinate representations of linear maps. The matrix is not the transformation itself: it depends on chosen bases in the domain and codomain. Once this distinction is kept visible, matrix multiplication, change of basis, rank and similarity all acquire transparent conceptual meanings. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 8

Determinants and Trace

The determinant compresses invertibility, oriented volume scaling and alternating multilinearity into one scalar. Trace measures the sum of diagonal action and is the simplest similarity invariant. Together they are the first examples of basis-independent quantities extracted from a matrix representation.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use alternating multilinear characterization in basis-free and coordinate form;
- explain and use permutation formula in basis-free and coordinate form;
- explain and use row operations in basis-free and coordinate form;
- explain and use invertibility in basis-free and coordinate form;
- explain and use geometric meaning in basis-free and coordinate form;
- explain and use trace in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

8.1 Alternating multilinear characterization

The determinant is the unique alternating multilinear function of the columns with $\det I = 1$.

8.2 Permutation formula

For $A = (a_{ij})$, $\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_i a_{i,\sigma(i)}$.

8.3 Row operations

Swapping rows changes sign, scaling a row scales the determinant, and adding a multiple of one row to another leaves it unchanged.

8.4 Invertibility

A square matrix is invertible exactly when its determinant is nonzero.

8.5 Geometric meaning

Over $\mathbb{R}$, $|\det A|$ is the factor by which the associated linear map scales n-dimensional volume; the sign records orientation.

8.6 Trace

$\operatorname{tr}(A) = \sum_i a_{ii}$ is linear and satisfies $\operatorname{tr}(AB) = \operatorname{tr}(BA)$, hence is invariant under similarity.

8.7 Core structural results

Theorem 8.1 (Multiplicativity of determinant)**.** *For square matrices A, B,* $\det(AB) = \det(A)\det(B)$*.*

Proof. For fixed B, the map sending the columns of A to $\det(AB)$ is alternating multilinear and has normalization $\det(B)$ at $A = I$; uniqueness gives the formula. $\square$

Theorem 8.2 (Cyclic trace identity)**.** *For compatible square matrices,* $\operatorname{tr}(AB) = \operatorname{tr}(BA)$*.*

Proof. Expand the diagonal entries: $\operatorname{tr}(AB) = \sum_{i,j} a_{ij} b_{ji}$ and interchange the dummy indices. $\square$

8.8 Worked examples

Example 8.3 (Triangular matrices)**.** The determinant of a triangular matrix is the product of its diagonal entries, and its trace is their sum.

Example 8.4 (Area scaling)**.** The matrix $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ scales signed area by 6.

Example 8.5 (Shear)**.** A shear $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ has determinant 1 although it changes angles.

8.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

8.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 8.6 (Determinant by elimination)**.** Compute $\det \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 5 \end{pmatrix}$ and explain why no expansion is needed.

Solution. The matrix is upper triangular, so the determinant is the product of diagonal entries: $1 \cdot 1 \cdot 5 = 5$. □

Problem 8.7 (Effect of a row replacement)**.** What happens to the determinant when row 2 is replaced by row $2 + 7$ row 1?

Solution. The determinant is unchanged. Multilinearity expands the new determinant into the old one plus 7 times a determinant with two equal rows, and the latter is zero by alternation. □

Problem 8.8 (Area scaling)**.** What signed area factor is produced by $A = \begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$?

Solution. $\det A = 6 - 1 = 5$. Thus oriented area is multiplied by 5; ordinary area is multiplied by $|\det A| = 5$. □

Problem 8.9 (Invertibility test)**.** Why does $\det A = 0$ imply the columns are dependent?

Solution. The determinant is alternating multilinear and vanishes on dependent columns. Conversely, if the columns were independent they would form a basis and the associated linear map would be invertible, forcing nonzero determinant. □

Problem 8.10 (Trace cyclicity)**.** Verify directly that $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Solution. $\operatorname{tr}(AB) = \sum_i \sum_j a_{ij} b_{ji}$. Interchanging the dummy indices gives $\sum_j \sum_i b_{ji} a_{ij} = \operatorname{tr}(BA)$. □

Problem 8.11 (Similarity invariants)**.** Show determinant and trace are invariant under $A \mapsto PAP^{-1}$.

Solution. Multiplicativity gives $\det(PAP^{-1}) = \det P \, \det A \, \det P^{-1} = \det A$. Cyclicity gives $\operatorname{tr}(PAP^{-1}) = \operatorname{tr}(AP^{-1}P) = \operatorname{tr} A$. □

Problem 8.12 (Determinant of a reflection)**.** Find the determinant of reflection across a hyperplane in $\mathbb{R}^n$.

Solution. The reflection fixes the $(n-1)$-dimensional hyperplane and negates one normal direction. In an adapted basis the matrix is $\operatorname{diag}(1, \ldots, 1, -1)$, so the determinant is -1. □

Problem 8.13 (Determinant of a shear)**.** Why does $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ preserve area for every t?

Solution. Its determinant is 1. Geometrically a shear slides parallel slices without changing base or height, consistent with area preservation. □

Problem 8.14 (Trace of rank one)**.** For column vectors u, v, compute $\operatorname{tr}(uv^T)$.

Solution. The diagonal entries are $u_i v_i$, so the trace is $\sum_i u_i v_i = v^T u$. ☐

Problem 8.15 (Block triangular determinant)**.** If $M = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix}$ with square diagonal blocks, what is $\det M$?

Solution. The determinant is $\det A \det C$. This follows by expansion or by reducing to triangular block operations; the off-diagonal block B does not affect the product of diagonal block determinants. ☐

Problem 8.16 (Orientation test)**.** How can determinant distinguish the two connected components of $GL_n(\mathbb{R})$?

Solution. An invertible real matrix has determinant either positive or negative. Continuous paths in GL_n cannot cross determinant 0, so the sign cannot change along a path. Positive and negative determinants correspond to the two orientation classes. ☐

Problem 8.17 (Determinant synthesis)**.** Why is the determinant better viewed as a property of an endomorphism than of one matrix?

Solution. Changing basis replaces the matrix by a similar one, and determinant is similarity invariant. Thus $\det T$ is well-defined independently of coordinates; a matrix merely computes it. ☐

8.11 Exercises with complete solutions

Exercise 8.18. Compute $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix}$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The permutation formula gives $ad - bc$. ☐

Exercise 8.19. Show that a matrix with two equal columns has determinant zero.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Alternation changes the determinant sign when the equal columns are swapped, but the matrix is unchanged, so the determinant equals its negative. ☐

Exercise 8.20. Why is $\det A \neq 0$ equivalent to invertibility?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Nonzero determinant implies full column independence; equivalently the associated endomorphism has zero kernel and hence is invertible. ☐

Exercise 8.21. Compute the determinant of a diagonal matrix.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Only the identity permutation contributes, so the determinant is the product of diagonal entries. ☐

Exercise 8.22. Prove $\operatorname{tr}(PAP^{-1}) = \operatorname{tr}(A)$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Use cyclicity: $\operatorname{tr}(PAP^{-1}) = \operatorname{tr}(AP^{-1}P) = \operatorname{tr}(A)$. ☐

Exercise 8.23. What determinant does a reflection of $\mathbb{R}^n$ across a hyperplane have?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. It has eigenvalues 1 on the hyperplane and -1 on a normal direction, so determinant -1. ☐

Exercise 8.24. Compute the trace of a rank-one matrix uv^T.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The trace is $v^T u = \sum_i u_i v_i$. ☐

Exercise 8.25. Show that determinant and trace are basis-independent for an endomorphism.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Two matrix representations are similar; determinant and trace are invariant under similarity. ☐

Chapter summary

The determinant compresses invertibility, oriented volume scaling and alternating multilinearity into one scalar. Trace measures the sum of diagonal action and is the simplest similarity invariant. Together they are the first examples of basis-independent quantities extracted from a matrix representation. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 9

Eigenvalues and Eigenvectors

Eigenvectors are directions preserved by an operator up to scaling, and eigenvalues are the corresponding scale factors. The characteristic polynomial packages all eigenvalues algebraically. This chapter emphasizes both geometric meaning and the limitations of eigenvector methods when a full eigenbasis does not exist.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use eigenvectors and eigenvalues in basis-free and coordinate form;
- explain and use eigenspaces in basis-free and coordinate form;
- explain and use characteristic polynomial in basis-free and coordinate form;
- explain and use algebraic and geometric multiplicity in basis-free and coordinate form;
- explain and use independence of eigenvectors in basis-free and coordinate form;
- explain and use field dependence in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

9.1 Eigenvectors and eigenvalues

A nonzero v is an eigenvector of T with eigenvalue λ when $T(v) = \lambda v$.

9.2 Eigenspaces

$E_\lambda = \ker(T - \lambda I)$ is a subspace. Distinct eigenspaces meet only at zero.

9.3 Characteristic polynomial

For a matrix A, $\chi_A(t) = \det(tI - A)$. Its roots are precisely the eigenvalues over the chosen field.

9.4 Algebraic and geometric multiplicity

The algebraic multiplicity is the multiplicity of a root of χ_A; the geometric multiplicity is $\dim E_\lambda$ and never exceeds it.

9.5 Independence of eigenvectors

Eigenvectors associated with distinct eigenvalues are linearly independent.

9.6 Field dependence

A real matrix may have no real eigenvalues but acquire complex eigenvalues after extending scalars to $\mathbb{C}$.

9.7 Core structural results

Theorem 9.1 (Distinct eigenvalues give independence)**.** *Eigenvectors corresponding to distinct eigenvalues are linearly independent.*

Proof. Use induction. Apply $T - \lambda_r I$ to a dependence relation; the last term vanishes and the remaining coefficients are multiplied by nonzero scalars. ☐

Theorem 9.2 (Geometric multiplicity bound)**.** *For an eigenvalue λ, $1 \leq \dim E_\lambda \leq$ its algebraic multiplicity.*

Proof. The lower bound is the definition of eigenvalue. The upper bound follows after choosing a basis beginning with an eigenspace basis, which forces a block with λ repeated on the diagonal. ☐

9.8 Worked examples

Example 9.3 (Diagonal matrix)**.** For $D = \operatorname{diag}(d_1, \ldots, d_n)$, each standard vector is an eigenvector with eigenvalue d_i.

Example 9.4 (Rotation)**.** A nontrivial planar rotation has no real eigenvectors unless the angle is 0 or π, but over $\mathbb{C}$ it has eigenvalues $e^{\pm i\theta}$.

Example 9.5 (Shear)**.** $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only the eigenvalue 1 and only a one-dimensional eigenspace.

9.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;

2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

9.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 9.6 (Defective shear)**.** For $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ compute algebraic and geometric multiplicities of $\lambda = 1$.

Solution. The characteristic polynomial is $(t-1)^2$, so algebraic multiplicity is 2. The eigenspace solves $(A - I)v = 0$, hence $y = 0$, so it is one-dimensional. Geometric multiplicity $1 < 2$, making the matrix defective. ☐

Problem 9.7 (Rotation and complexification)**.** Find the eigenvalues of the real rotation matrix $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ over $\mathbb{R}$ and $\mathbb{C}$.

Solution. The characteristic polynomial is $t^2 + 1$. It has no real roots, so no real eigenvalues; over $\mathbb{C}$ the eigenvalues are i and $-i$. ☐

Problem 9.8 (Parameter collision)**.** For $A_t = \begin{pmatrix} t & 1 \\ 0 & 2 \end{pmatrix}$, when are there two distinct eigenvalues?

Solution. The eigenvalues are the diagonal entries t and 2. They are distinct exactly when $t \neq 2$. At $t = 2$ the matrix becomes a nontrivial Jordan-type block with only one eigendirection. ☐

Problem 9.9 (Eigenvectors of powers)**.** If $Tv = \lambda v$, what is $T^k v$?

Solution. Induction gives $T^k v = \lambda^k v$. Thus eigenvectors of T remain eigenvectors of every polynomial in T, with eigenvalues transformed by the polynomial. ☐

Problem 9.10 (Zero eigenvalue and singularity)**.** Show that 0 is an eigenvalue of a square matrix iff the matrix is singular.

Solution. 0 is an eigenvalue iff there is $v \neq 0$ with $Av = 0$, i.e. iff the kernel is nontrivial. A square matrix is invertible exactly when its kernel is zero. ☐

Problem 9.11 (Distinct eigenvectors)**.** Why are eigenvectors for distinct eigenvalues independent?

Solution. In a dependence relation, apply $T - \lambda_r I$ to eliminate one eigenvector while multiplying the others by nonzero scalars. Induction reduces to a shorter relation, forcing all coefficients to zero. ☐

Problem 9.12 (Triangular spectrum)**.** Read the eigenvalues of $\begin{pmatrix} 3 & * & * \\ 0 & -1 & * \\ 0 & 0 & 3 \end{pmatrix}$.

Solution. The characteristic polynomial of a triangular matrix is the product $(t-3)(t+1)(t-3)$, so the eigenvalues are 3 with algebraic multiplicity 2 and -1 with multiplicity 1. ☐

Problem 9.13 (Eigenspace as kernel)**.** Why is E_λ always a subspace?

Solution. $E_\lambda = \ker(T - \lambda I)$. Kernels of linear maps are subspaces, so closure and the zero vector follow automatically. ☐

Problem 9.14 (Eigenvalue of an inverse)**.** If T is invertible and $Tv = \lambda v$, what eigenvalue does v have for T^{-1}?

Solution. Invertibility implies $\lambda \neq 0$. Applying T^{-1} gives $v = \lambda T^{-1}v$, so $T^{-1}v = \lambda^{-1}v$. ☐

Problem 9.15 (Trace and eigenvalues)**.** For a triangularizable operator, relate trace to eigenvalues counted with algebraic multiplicity.

Solution. In a triangular representation the diagonal entries are the eigenvalues with multiplicity, and trace is the sum of diagonal entries. Since trace is basis-invariant, it equals the eigenvalue sum. ☐

Problem 9.16 (Determinant and eigenvalues)**.** For a triangularizable operator, relate determinant to eigenvalues.

Solution. In triangular form the determinant is the product of diagonal entries, which are the eigenvalues with algebraic multiplicity. Therefore $\det T$ equals their product. ☐

Problem 9.17 (Spectral synthesis)**.** Why can knowing all eigenvalues still be insufficient to determine an operator up to similarity?

Solution. Eigenvalues record only the roots of the characteristic polynomial. Operators with the same eigenvalues can have different eigenspace dimensions or Jordan block structures, such as the identity and a nontrivial shear with eigenvalue 1. ☐

9.11 Exercises with complete solutions

Exercise 9.18. Find the eigenvalues of $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. They are the diagonal entries 2 and 3. ☐

Exercise 9.19. Find the eigenspace for eigenvalue 1 of $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Solve $(A - I)v = 0$, giving $y = 0$; the eigenspace is the span of $(1, 0)$. ☐

Exercise 9.20. Why is the zero vector excluded from the definition of eigenvector?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. It satisfies $T(0) = \lambda 0$ for every scalar, so including it would carry no information about the operator. □

Exercise 9.21. Prove eigenspaces are subspaces.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. They are kernels of the linear maps $T - \lambda I$. □

Exercise 9.22. Show that eigenvalues depend on the scalar field.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The real rotation by 90° has characteristic polynomial $t^2 + 1$, with no real roots but roots $\pm i$ over $\mathbb{C}$. □

Exercise 9.23. If T is invertible, can 0 be an eigenvalue?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. No. A zero eigenvalue would give a nonzero vector in the kernel. □

Exercise 9.24. What are the eigenvalues of a triangular matrix?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. They are its diagonal entries, because the characteristic polynomial is the product of $t - a_{ii}$. □

Exercise 9.25. Explain the difference between algebraic and geometric multiplicity for a shear.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. For the 2×2 shear, eigenvalue 1 has algebraic multiplicity 2 but geometric multiplicity 1. □

Chapter summary

Eigenvectors are directions preserved by an operator up to scaling, and eigenvalues are the corresponding scale factors. The characteristic polynomial packages all eigenvalues algebraically. This chapter emphasizes both geometric meaning and the limitations of eigenvector methods when a full eigenbasis does not exist. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 10

Invariant Subspaces and Triangularization

Invariant subspaces let a large operator be studied through smaller pieces. Triangularization is the systematic expression of this idea: once an invariant flag is chosen, the matrix becomes triangular and its diagonal reveals the eigenvalues. Over an algebraically closed field, every operator on a nonzero finite-dimensional space admits such a form.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use invariant subspaces in basis-free and coordinate form;
- explain and use quotient operators in basis-free and coordinate form;
- explain and use invariant flags in basis-free and coordinate form;
- explain and use triangular matrices and flags in basis-free and coordinate form;
- explain and use triangularization theorem in basis-free and coordinate form;
- explain and use block triangular form in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

10.1 Invariant subspaces

A subspace U is T-invariant when $T(U) \subseteq U$. The restriction $T|_U$ is then an operator on U.

10.2 Quotient operators

If U is invariant, T induces an operator on V/U by $v + U \mapsto T(v) + U$.

10.3 Invariant flags

A chain $0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$ with $\dim V_k = k$ and each V_k invariant is a complete invariant flag.

10.4 Triangular matrices and flags

An operator is upper triangular in a basis exactly when the spans of its initial basis segments form an invariant flag.

10.5 Triangularization theorem

Over an algebraically closed field, every operator on a finite-dimensional nonzero vector space is triangularizable.

10.6 Block triangular form

Even without a complete flag, a single invariant subspace produces a block upper-triangular matrix, separating the restriction and quotient actions.

10.7 Core structural results

Theorem 10.1 (Flag criterion for triangularity)**.** *A basis $v_1, \ldots, v_n$ gives an upper-triangular matrix for T if and only if* $\operatorname{span}(v_1, \ldots, v_k)$ *is invariant for every k.*

Proof. Upper triangularity says $T(v_k)$ lies in the span of the first k basis vectors. This is exactly the invariant-flag condition. ☐

Theorem 10.2 (Triangularization over an algebraically closed field)**.** *Every operator on a finite-dimensional complex vector space has an upper-triangular matrix in some basis.*

Proof. Choose an eigenvector, use its span as the first invariant subspace, pass to the quotient, and induct on dimension. ☐

10.8 Worked examples

Example 10.3 (Nilpotent shift)**.** The map $N(e_1) = 0$, $N(e_k) = e_{k-1}$ preserves the standard flag and is strictly upper triangular in the reversed standard convention.

Example 10.4 (Polynomial operator)**.** Differentiation preserves the nested polynomial spaces $\mathcal{P}_0 \subset \mathcal{P}_1 \subset \cdots$.

Example 10.5 (Real rotation)**.** A planar rotation through 90° has no invariant real line, so it cannot be triangularized over $\mathbb{R}$.

10.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;

2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

10.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 10.6 (Invariant line)**.** For $A = \begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$, show $\operatorname{span}(e_1)$ is invariant.

Solution. $Ae_1 = 2e_1$, so every scalar multiple of e_1 is mapped back into the same line. It is in fact an eigenspace. ☐

Problem 10.7 (Block form from invariance)**.** If U is invariant under T, why does a basis beginning with a basis of U give a block upper-triangular matrix?

Solution. Images of basis vectors in U stay in U, so their coordinate columns have zero components in the complementary basis directions. That forces the lower-left block to vanish. ☐

Problem 10.8 (Quotient operator)**.** Define the induced operator on V/U when U is T-invariant and prove it is well-defined.

Solution. Set $\overline{T}(v + U) = T(v) + U$. If $v - w \in U$, invariance gives $T(v - w) \in U$, so $T(v) + U = T(w) + U$. Thus the definition is independent of representative. ☐

Problem 10.9 (Flag criterion)**.** Explain why an upper-triangular matrix preserves the coordinate flag $V_k = \operatorname{span}(e_1, \dots, e_k)$.

Solution. The first k columns of an upper-triangular matrix have zeros below row k, so the image of any vector supported in the first k coordinates remains supported there. ☐

Problem 10.10 (Triangularization over $\mathbb{C}$)**.** Outline the induction proving every complex operator is triangularizable.

Solution. Choose an eigenvector and its invariant line. Pass to the quotient, which has one lower dimension, triangularize the induced operator there by induction, then lift the quotient basis and prepend the eigenvector. ☐

Problem 10.11 (Failure over $\mathbb{R}$)**.** Why is rotation by 90° not triangularizable over $\mathbb{R}$?

Solution. A real upper-triangular matrix has real diagonal entries and hence real eigenvalues. The rotation has characteristic polynomial $t^2 + 1$ and no real eigenvalue, so no real triangular form exists. ☐

Problem 10.12 (Polynomial invariant subspaces)**.** Show $\mathcal{P}_k$ is invariant under differentiation on $\mathcal{P}_n$.

Solution. If $\deg p \le k$, then $\deg p' \le k - 1 \le k$ (or $p' = 0$). Hence differentiation maps $\mathcal{P}_k$ into itself. ☐

Problem 10.13 (Invariant complement need not exist)**.** Give a 2×2 example with an invariant line but no invariant complementary line.

Solution. The shear $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ preserves $\mathrm{span}(e_1)$. Its only eigenspace is that line, so no distinct invariant line can complement it. ☐

Problem 10.14 (Restriction spectrum)**.** If U is invariant, why must every eigenvalue of $T|_U$ also be an eigenvalue of T?

Solution. An eigenvector of the restriction is a nonzero vector $u \in U$ with $T(u) = \lambda u$, which is also an eigenvector of T on the whole space. ☐

Problem 10.15 (Nested kernels)**.** For a nilpotent N, show $\ker N \subseteq \ker N^2 \subseteq \cdots$ are invariant.

Solution. If $N^k v = 0$, then $N^k(Nv) = N^{k+1}v = 0$, so $Nv \in \ker N^k$ after adjusting indices appropriately; each kernel is invariant under N. ☐

Problem 10.16 (Diagonal form as strongest triangular form)**.** How does diagonalization strengthen triangularization?

Solution. Triangularization produces a complete invariant flag but can retain couplings above the diagonal. Diagonalization decomposes the space into one-dimensional invariant eigenspaces, eliminating all off-diagonal couplings. ☐

Problem 10.17 (Invariant-subspace synthesis)**.** Why are invariant subspaces the natural mechanism for induction in operator theory?

Solution. They let one split the problem into a restriction on a smaller subspace and an induced operator on a lower-dimensional quotient. Structural information can then be assembled from these two simpler pieces. ☐

10.11 Exercises with complete solutions

Exercise 10.18. Show every eigenspace is invariant.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. If $v \in E_\lambda$, then $T(v) = \lambda v \in E_\lambda$. ☐

Exercise 10.19. If U is invariant, prove the quotient operator is well-defined.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. If $v + U = w + U$, then $v - w \in U$, so $T(v - w) \in U$ and $T(v) + U = T(w) + U$. ☐

Exercise 10.20. Why does an upper-triangular matrix preserve the coordinate flag?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The first k columns have zeros below row k, so vectors supported in the first k coordinates map back into that subspace. ☐

Exercise 10.21. Give an invariant subspace of a diagonal operator.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The span of any subset of eigenbasis vectors is invariant. □

Exercise 10.22. Why may triangularization fail over $\mathbb{R}$?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The induction starts by choosing an eigenvector; a real operator can have no real eigenvalue. □

Exercise 10.23. Show the characteristic polynomial of a triangular matrix is the product of $t - a_{ii}$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The matrix $tI - A$ is triangular, so its determinant is the product of its diagonal entries. □

Exercise 10.24. Describe the block form associated with an invariant subspace U.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Choose a basis of U and extend it to V. The lower-left block is zero because $T(U) \subseteq U$. □

Exercise 10.25. Explain why triangularization already reveals all eigenvalues.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The characteristic polynomial is the product of t minus the diagonal entries, so the diagonal entries are precisely the eigenvalues with multiplicity. □

Chapter summary

Invariant subspaces let a large operator be studied through smaller pieces. Triangularization is the systematic expression of this idea: once an invariant flag is chosen, the matrix becomes triangular and its diagonal reveals the eigenvalues. Over an algebraically closed field, every operator on a nonzero finite-dimensional space admits such a form. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 11

Diagonalization and Minimal Polynomials

Diagonalization asks whether an operator can be decomposed into independent one-dimensional modes. The minimal polynomial gives the sharp algebraic criterion: an operator is diagonalizable exactly when its minimal polynomial splits into distinct linear factors. This viewpoint unifies eigenvalue counting with polynomial identities satisfied by the operator.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use diagonalizable operators in basis-free and coordinate form;
- explain and use direct sums of eigenspaces in basis-free and coordinate form;
- explain and use minimal polynomial in basis-free and coordinate form;
- explain and use divisibility property in basis-free and coordinate form;
- explain and use square-free criterion in basis-free and coordinate form;
- explain and use spectral projectors by interpolation in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

11.1 Diagonalizable operators

An operator is diagonalizable when it has a basis of eigenvectors, equivalently when some matrix representation is diagonal.

11.2 Direct sums of eigenspaces

If the characteristic polynomial splits, diagonalizability is equivalent to V being the direct sum of its eigenspaces.

11.3 Minimal polynomial

The minimal polynomial m_T is the monic polynomial of least positive degree with $m_T(T) = 0$.

11.4 Divisibility property

Every polynomial p with $p(T) = 0$ is divisible by m_T. In particular, m_T divides the characteristic polynomial.

11.5 Square-free criterion

If m_T splits, then T is diagonalizable exactly when m_T has no repeated root.

11.6 Spectral projectors by interpolation

For distinct eigenvalues, Lagrange interpolation polynomials in T produce projections onto eigenspaces.

11.7 Core structural results

Theorem 11.1 (Dimension criterion). *If χ_T splits, then T is diagonalizable if and only if the sum of eigenspace dimensions equals* $\dim V$*.*

Proof. Distinct eigenspaces form a direct sum. Their union of bases is an eigenbasis exactly when the direct sum has full dimension. ☐

Theorem 11.2 (Minimal-polynomial criterion). *An operator is diagonalizable if and only if its minimal polynomial is a product of distinct linear factors.*

Proof. For a diagonal operator, a polynomial annihilates it exactly when it vanishes at every diagonal eigenvalue. Conversely, use Lagrange polynomials to construct complementary eigenspace projectors. ☐

11.8 Worked examples

Example 11.3 (Projection). A projection $P^2 = P$ has minimal polynomial dividing $t(t-1)$, which has distinct roots, so every projection is diagonalizable.

Example 11.4 (Involution). If $T^2 = I$ over characteristic not 2, then m_T divides $(t-1)(t+1)$, so T is diagonalizable.

Example 11.5 (Nontrivial shear). The shear has minimal polynomial $(t-1)^2$, so it is not diagonalizable.

11.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

11.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 11.6 (Projection minimal polynomial)**.** If $P^2 = P$ and $P \neq 0, I$, determine the minimal polynomial.

Solution. P satisfies $t(t-1)$, and neither t nor $t-1$ alone annihilates it because P is neither zero nor identity. Thus $m_P(t) = t(t-1)$, which has distinct roots, so P is diagonalizable. □

Problem 11.7 (Involution)**.** If $T^2 = I$ over a field of characteristic not 2, prove T is diagonalizable.

Solution. The minimal polynomial divides $(t-1)(t+1)$, a product of distinct linear factors. Therefore the minimal polynomial is square-free and split, which is the diagonalizability criterion. □

Problem 11.8 (Shear obstruction)**.** Why is a nontrivial shear not diagonalizable?

Solution. For $A = I + N$ with $N \neq 0$ and $N^2 = 0$, the minimal polynomial is $(t-1)^2$. The repeated factor violates the square-free minimal-polynomial criterion. □

Problem 11.9 (Eigenspace dimension test)**.** An operator on a 5-dimensional space has eigenspaces of dimensions $2, 1, 2$ for three distinct eigenvalues. Is it diagonalizable?

Solution. Distinct eigenspaces form a direct sum, and their dimensions sum to 5. Their combined bases therefore form a basis of eigenvectors, so the operator is diagonalizable. □

Problem 11.10 (Minimal polynomial from diagonal data)**.** For $D = \mathrm{diag}(1, 1, 3, 5, 5)$ find m_D.

Solution. The minimal polynomial must vanish at each distinct diagonal value and needs no repeated factors: $m_D = (t-1)(t-3)(t-5)$. □

Problem 11.11 (Annihilating polynomial divisibility)**.** Why does the minimal polynomial divide every polynomial p with $p(T) = 0$?

Solution. Divide $p = qm_T + r$ with $\deg r < \deg m_T$. Evaluating at T gives $0 = r(T)$. Minimality of m_T forces $r = 0$, so $m_T \mid p$. □

Problem 11.12 (Spectral projectors)**.** For distinct eigenvalues λ_1, λ_2, write the projector onto E_{λ_1} as a polynomial in T.

Solution. $P_1 = (T - \lambda_2 I)/(\lambda_1 - \lambda_2)$. It acts as 1 on E_{λ_1} and 0 on E_{λ_2}. □

Problem 11.13 (Repeated characteristic root)**.** Why does a repeated root of the characteristic polynomial not imply non-diagonalizability?

Solution. The identity has characteristic polynomial $(t-1)^n$ but minimal polynomial $t-1$. Diagonalizability is controlled by repeated roots in the minimal polynomial, not merely in the characteristic polynomial. ☐

Problem 11.14 (Direct eigenspace sum)**.** Prove eigenspaces for distinct eigenvalues have zero intersection.

Solution. If v lies in both E_λ and E_μ with $\lambda \neq \mu$, then $\lambda v = Tv = \mu v$, so $(\lambda - \mu)v = 0$ and hence $v = 0$. ☐

Problem 11.15 (Diagonalization by eigenbasis)**.** Given n independent eigenvectors in an n-dimensional space, how do you diagonalize the operator?

Solution. Use the eigenvectors as an ordered basis. The image of each basis vector is its eigenvalue times itself, so the representing matrix is diagonal with those eigenvalues. ☐

Problem 11.16 (Polynomial functional test)**.** If T is diagonalizable, why is every polynomial $p(T)$ diagonalizable in the same basis?

Solution. In an eigenbasis, T is diagonal. Applying a polynomial entrywise to the diagonal gives another diagonal matrix, so $p(T)$ is diagonal in the same basis. ☐

Problem 11.17 (Diagonalization synthesis)**.** Compare the roles of characteristic and minimal polynomials in diagonalization.

Solution. The characteristic polynomial tells which eigenvalues are available and their algebraic multiplicities. The minimal polynomial records the largest Jordan-type obstruction. Splitting of the characteristic polynomial is necessary, while square-freeness of the split minimal polynomial is the decisive criterion. ☐

11.11 Exercises with complete solutions

Exercise 11.18. Why is every operator with distinct eigenvalues equal in number to the dimension diagonalizable?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Eigenvectors for distinct eigenvalues are independent, yielding a basis. ☐

Exercise 11.19. Find the minimal polynomial of a diagonal matrix with diagonal entries $1, 1, 3$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. It is $(t-1)(t-3)$; repeated diagonal occurrences do not create repeated factors in the minimal polynomial. ☐

Exercise 11.20. Find the minimal polynomial of a nontrivial 2×2 shear.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. It is $(t-1)^2$: $A-I$ is nonzero but $(A-I)^2=0$. □

Exercise 11.21. Show every projection is diagonalizable without computing a determinant.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Decompose $V = \operatorname{im} P \oplus \ker P$. The operator acts as 1 on the image and 0 on the kernel. □

Exercise 11.22. If $T^2 = I$, what are the possible eigenvalues?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Any eigenvalue λ satisfies $\lambda^2 = 1$, so $\lambda = \pm 1$. □

Exercise 11.23. Why must m_T divide every annihilating polynomial?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Divide p by m_T; the remainder has smaller degree and also annihilates T, so minimality forces the remainder to vanish. □

Exercise 11.24. Construct the spectral projector onto E_{λ_i} for distinct eigenvalues.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Use $P_i = \prod_{j\neq i}(T-\lambda_j I)/(\lambda_i-\lambda_j)$; it equals 1 on E_{λ_i} and 0 on the other eigenspaces. □

Exercise 11.25. Explain why repeated roots in the characteristic polynomial do not by themselves prevent diagonalization.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The identity matrix has characteristic polynomial $(t-1)^n$ but is diagonal; what matters is the minimal polynomial, which is $t-1$. □

Chapter summary

Diagonalization asks whether an operator can be decomposed into independent one-dimensional modes. The minimal polynomial gives the sharp algebraic criterion: an operator is diagonalizable exactly when its minimal polynomial splits into distinct linear factors. This viewpoint unifies eigenvalue counting with polynomial identities satisfied by the operator. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 12

Canonical Forms

When diagonalization fails, canonical forms measure the failure rather than hiding it. Jordan form over an algebraically closed field records nilpotent chains attached to each eigenvalue, while rational canonical form works over arbitrary fields using invariant factors. Both arise from viewing a vector space as a module over the polynomial ring via $p \cdot v = p(T)v$.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use generalized eigenvectors in basis-free and coordinate form;
- explain and use nilpotent operators in basis-free and coordinate form;
- explain and use jordan blocks in basis-free and coordinate form;
- explain and use jordan canonical form in basis-free and coordinate form;
- explain and use primary decomposition in basis-free and coordinate form;
- explain and use rational canonical form in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

12.1 Generalized eigenvectors

A vector lies in the generalized eigenspace for λ if $(T - \lambda I)^k v = 0$ for some k.

12.2 Nilpotent operators

Nilpotent operators organize into chains $v, Nv, \ldots$. Their Jordan blocks encode chain lengths.

12.3 Jordan blocks

A Jordan block $J_r(\lambda)$ has λ on the diagonal and ones on the superdiagonal. Its minimal polynomial is $(t-\lambda)^r$.

12.4 Jordan canonical form

Over an algebraically closed field, every operator is similar to a direct sum of Jordan blocks, unique up to block order.

12.5 Primary decomposition

If the minimal polynomial factors into pairwise coprime powers, the space decomposes into invariant primary components.

12.6 Rational canonical form

Over any field, invariant factors determine a block diagonal companion-matrix form. Jordan form is the split refinement of this module-theoretic structure.

12.7 Core structural results

Theorem 12.1 (Generalized eigenspace decomposition)**.** *If $m_T(t) = \prod_i (t-\lambda_i)^{r_i}$ with distinct λ_i, then $V = \bigoplus_i \ker(T-\lambda_i I)^{r_i}$.*

Proof. The factors are pairwise coprime. Bézout identities in $\mathbb{F}[t]$ produce polynomial projectors whose images are the primary components. ☐

Theorem 12.2 (Jordan block invariants)**.** *For a nilpotent operator, the numbers $\dim\ker N^k$ determine the Jordan block sizes.*

Proof. The increment $\dim\ker N^k - \dim\ker N^{k-1}$ counts blocks of size at least k; successive differences recover the exact sizes. ☐

12.8 Worked examples

Example 12.3 (Single Jordan block)**.** $J_3(0)$ is nilpotent of index 3; kernel dimensions are $1, 2, 3$.

Example 12.4 (Two blocks)**.** $J_2(0) \oplus J_1(0)$ has kernel dimensions $2, 3$, revealing two blocks, one of which has length at least 2.

Example 12.5 (Companion matrix)**.** The companion matrix of a monic polynomial has that polynomial as both characteristic and minimal polynomial when it represents a cyclic module.

12.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;

2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

12.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 12.6 (Nilpotent chain)**.** For $N = J_3(0)$ compute $\dim\ker N$, $\dim\ker N^2$, and $\dim\ker N^3$.

Solution. The standard Jordan chain has lengths three. The kernels have dimensions $1, 2, 3$ respectively. The increments $1, 1, 1$ indicate one block of size at least $1, 2, 3$. ☐

Problem 12.7 (Two nilpotent blocks)**.** If a nilpotent operator has Jordan blocks of sizes 3 and 1, compute $\dim\ker N^k$ for $k = 1, 2, 3$.

Solution. At $k = 1$, each block contributes one kernel dimension, giving 2. At $k = 2$, the size-3 block contributes 2 and the size-1 block 1, giving 3. At $k = 3$ the total is 4. ☐

Problem 12.8 (Geometric multiplicity in Jordan form)**.** How is the geometric multiplicity of λ read from Jordan form?

Solution. Each Jordan block for λ contributes exactly one eigenvector direction, so the geometric multiplicity is the number of Jordan blocks carrying λ. ☐

Problem 12.9 (Minimal polynomial from blocks)**.** If the largest Jordan block for eigenvalue 2 has size 4 and for eigenvalue -1 has size 2, find the minimal polynomial.

Solution. The exponents are the largest block sizes, so $m_T(t) = (t-2)^4(t+1)^2$. ☐

Problem 12.10 (Diagonalizability from blocks)**.** When is a Jordan form diagonal?

Solution. Exactly when every Jordan block has size 1. Equivalently every eigenvalue's generalized eigenspace is already its ordinary eigenspace. ☐

Problem 12.11 (Generalized eigenspace)**.** Show $\ker(T - \lambda I)^k$ is T-invariant.

Solution. T commutes with every polynomial in T, so $(T - \lambda I)^k T(v) = T(T - \lambda I)^k v$. If the latter is zero, Tv lies in the same kernel. ☐

Problem 12.12 (Primary decomposition idea)**.** Why do coprime factors of the minimal polynomial lead to a direct-sum decomposition?

Solution. Bézout identities between the coprime polynomial factors produce polynomial projectors in T whose images are the corresponding primary kernels and whose sum is the identity. ☐

Problem 12.13 (Real versus complex canonical form)**.** Why may Jordan form fail over $\mathbb{R}$?

Solution. Jordan form requires the minimal polynomial to split into linear factors. A real operator can have irreducible quadratic factors, such as a planar rotation. Rational canonical form does not require splitting. ☐

Problem 12.14 (Companion matrix)**.** What is the conceptual role of a companion matrix?

Solution. It represents multiplication by t on the cyclic module $\mathbb{F}[t]/(p)$. Its characteristic and minimal polynomial are both the defining monic polynomial p when the module is cyclic. □

Problem 12.15 (Kernel increments)**.** How do the increments $d_k = \dim\ker N^k - \dim\ker N^{k-1}$ encode block sizes?

Solution. d_k counts the number of Jordan blocks of size at least k. Therefore $d_k - d_{k+1}$ counts blocks of size exactly k. □

Problem 12.16 (Similarity invariants)**.** Name data preserved by similarity that are visible in Jordan form.

Solution. Eigenvalues, algebraic and geometric multiplicities, minimal polynomial, and the multiset of Jordan block sizes are all similarity invariants. Jordan form packages them completely over an algebraically closed field. □

Problem 12.17 (Canonical-form synthesis)**.** Why is the module viewpoint $p \cdot v = p(T)v$ natural?

Solution. It turns invariant subspaces into submodules and polynomial identities into annihilators. The structure theorem for finitely generated modules over $\mathbb{F}[t]$ then yields rational canonical form, with Jordan form as the split primary refinement. □

12.11 Exercises with complete solutions

Exercise 12.18. Compute the minimal polynomial of $J_4(\lambda)$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. It is $(t-\lambda)^4$ because the nilpotent part has index 4. □

Exercise 12.19. What is the geometric multiplicity of an eigenvalue in Jordan form?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. It equals the number of Jordan blocks carrying that eigenvalue. □

Exercise 12.20. How can kernel dimensions reveal nilpotent block sizes?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The increment from $\ker N^{k-1}$ to $\ker N^k$ counts blocks of size at least k. □

Exercise 12.21. When is a Jordan matrix diagonal?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Exactly when every Jordan block has size 1. □

Exercise 12.22. Why is Jordan form not always available over $\mathbb{R}$?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. The characteristic/minimal polynomial may not split into linear factors over $\mathbb{R}$. □

Exercise 12.23. State the advantage of rational canonical form.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. It exists over every field and does not require the characteristic polynomial to split. □

Exercise 12.24. Show that generalized eigenspaces are invariant.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. If $(T-\lambda I)^k v = 0$, then $(T-\lambda I)^k T(v) = T(T-\lambda I)^k v = 0$. □

Exercise 12.25. Relate the largest Jordan block for λ to the minimal polynomial.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Its size is the exponent of $(t-\lambda)$ in the minimal polynomial. □

Chapter summary

When diagonalization fails, canonical forms measure the failure rather than hiding it. Jordan form over an algebraically closed field records nilpotent chains attached to each eigenvalue, while rational canonical form works over arbitrary fields using invariant factors. Both arise from viewing a vector space as a module over the polynomial ring via $p \cdot v = p(T)v$. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Part III

Euclidean and Hilbert-Space Geometry

Chapter 13

Inner Products and Orthogonality

Inner products add geometry to linear algebra. Length, angle, orthogonality and best approximation all come from a positive definite sesquilinear form. The complex case forces careful placement of conjugation, but the resulting theory unifies Euclidean geometry with finite-dimensional Hilbert-space methods.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use inner products in basis-free and coordinate form;
- explain and use norm and distance in basis-free and coordinate form;
- explain and use cauchy–schwarz in basis-free and coordinate form;
- explain and use orthogonality in basis-free and coordinate form;
- explain and use orthogonal complements in basis-free and coordinate form;
- explain and use parallelogram and polarization in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

13.1 Inner products

A real inner product is symmetric bilinear and positive definite; a complex inner product is conjugate symmetric and sesquilinear.

13.2 Norm and distance

The norm is $\|v\| = \sqrt{\langle v, v\rangle}$ and the induced distance is $\|v - w\|$.

13.3 Cauchy–Schwarz

$|\langle u, v\rangle| \leq \|u\|\|v\|$, with equality precisely when the vectors are linearly dependent.

13.4 Orthogonality

Vectors are orthogonal when their inner product is zero. Orthogonal nonzero lists are automatically independent.

13.5 Orthogonal complements

For a subspace U, $U^\perp = \{v : \langle v, u\rangle = 0 \ \forall u \in U\}$. In finite dimensions, $V = U \oplus U^\perp$.

13.6 Parallelogram and polarization

The inner product determines the norm, and conversely a norm comes from an inner product exactly when it satisfies the parallelogram law.

13.7 Core structural results

Theorem 13.1 (Cauchy–Schwarz inequality)**.** *For all u, v in an inner-product space, $|\langle u, v\rangle| \leq \|u\|\|v\|$.*

Proof. If $v \neq 0$, apply positivity to $u - (\langle u, v\rangle/\|v\|^2)v$ and expand. The case $v = 0$ is immediate. □

Theorem 13.2 (Pythagorean theorem)**.** *If $u \perp v$, then $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.*

Proof. Expand $\langle u + v, u + v\rangle$ and use conjugate symmetry together with the vanishing cross terms. □

13.8 Worked examples

Example 13.3 (Euclidean product)**.** On $\mathbb{R}^n$, $\langle x, y\rangle = x^T y$.

Example 13.4 (Weighted product)**.** For positive weights w_i, $\langle x, y\rangle = \sum_i w_i x_i y_i$ is an inner product.

Example 13.5 (Function product)**.** On continuous functions, $\langle f, g\rangle = \int_0^1 f(t)\overline{g(t)}\,dt$ is an inner product.

13.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

13.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 13.6 (Cauchy–Schwarz equality)**.** When does equality hold in Cauchy–Schwarz?

Solution. Equality holds exactly when one vector is a scalar multiple of the other. In the standard proof, equality means the orthogonal residual $u - (\langle u, v\rangle/\|v\|^2)v$ has norm zero. □

Problem 13.7 (Orthogonal independence)**.** Prove nonzero pairwise orthogonal vectors are linearly independent.

Solution. Take $\sum a_i v_i = 0$ and inner-product with v_j. All cross terms vanish, leaving $a_j\|v_j\|^2 = 0$. Since $v_j \neq 0$, $a_j = 0$ for every j. □

Problem 13.8 (Weighted inner product)**.** Show $\langle x, y\rangle = 2x_1y_1 + 5x_2y_2$ is an inner product on $\mathbb{R}^2$.

Solution. Bilinearity and symmetry are immediate. Positive definiteness follows because $\langle x, x\rangle = 2x_1^2 + 5x_2^2 > 0$ for every nonzero x. □

Problem 13.9 (Orthogonal complement)**.** Find the orthogonal complement of $\operatorname{span}(1, 1, 1)$ in $\mathbb{R}^3$.

Solution. It is the plane $\{(x, y, z) : x + y + z = 0\}$, since orthogonality to $(1, 1, 1)$ is exactly the displayed equation. □

Problem 13.10 (Pythagoras)**.** If $u \perp v$, prove $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.

Solution. Expand $\langle u + v, u + v\rangle$. The cross terms vanish by orthogonality, leaving the sum of the squared norms. □

Problem 13.11 (Triangle inequality)**.** Derive the triangle inequality from Cauchy–Schwarz.

Solution. $\|u + v\|^2 = \|u\|^2 + 2\operatorname{Re}\langle u, v\rangle + \|v\|^2 \leq (\|u\| + \|v\|)^2$. Taking nonnegative square roots yields the result. □

Problem 13.12 (Complex conjugation placement)**.** Why must the standard complex inner product include conjugation?

Solution. Without conjugation, $\sum z_i^2$ can be nonreal or negative-looking; for example $i^2 = -1$. With conjugation, $\langle z, z\rangle = \sum |z_i|^2 \geq 0$ and is zero only for $z = 0$. □

Problem 13.13 (Distance to a subspace)**.** If $V = U \oplus U^\perp$ and $v = u + w$, which component controls the distance from v to U?

Solution. For any $u' \in U$, $v - u' = (u - u') + w$ with orthogonal terms, so $\|v - u'\|^2 = \|u - u'\|^2 + \|w\|^2$. The minimal distance is $\|w\|$. □

Problem 13.14 (Polarization)**.** Recover a real inner product from its norm.

Solution. The real polarization identity is $\langle u, v\rangle = \frac{1}{4}(\|u + v\|^2 - \|u - v\|^2)$. Expanding the two norms verifies the formula. □

Problem 13.15 (Angle computation)**.** Find the angle between $(1, 1, 0)$ and $(1, 0, 1)$.

Solution. The dot product is 1, each norm is $\sqrt{2}$, so $\cos\theta = 1/2$ and $\theta = \pi/3$. □

Problem 13.16 (Orthogonal decomposition dimension)**.** If U is a subspace of finite-dimensional V, prove $\dim U + \dim U^{\perp} = \dim V$.

Solution. Choose an orthonormal basis of U and extend it to an orthonormal basis of V. The added basis vectors form a basis of $U^{\perp}$, so the dimensions add. ☐

Problem 13.17 (Inner-product synthesis)**.** Why is an inner product more structure than a vector-space norm?

Solution. An inner product determines a norm, but not every norm comes from an inner product. The parallelogram identity characterizes inner-product norms. Thus orthogonality and angle require extra geometric structure beyond linear operations. ☐

13.11 Exercises with complete solutions

Exercise 13.18. Show nonzero orthogonal vectors are independent.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Take a dependence relation and inner-product with one vector; all cross terms vanish, leaving its coefficient times a positive norm squared. ☐

Exercise 13.19. Compute the angle between $(1, 0)$ and $(1, 1)$ in $\mathbb{R}^2$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The cosine is $1/(1 \cdot \sqrt{2}) = 1/\sqrt{2}$, so the angle is $\pi/4$. ☐

Exercise 13.20. Find the orthogonal complement of the x-axis in $\mathbb{R}^2$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. It is the y-axis. ☐

Exercise 13.21. Verify the parallelogram identity.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Expand $\|u + v\|^2 + \|u - v\|^2$; the cross terms cancel, leaving $2\|u\|^2 + 2\|v\|^2$. ☐

Exercise 13.22. Why is complex conjugation necessary in the standard complex inner product?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Without conjugation, $\langle v, v\rangle$ need not be real or positive; conjugation gives $\sum |v_i|^2$. ☐

Exercise 13.23. Show $U \cap U^{\perp} = \{0\}$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. If v lies in both, then $\langle v, v\rangle = 0$, hence $v = 0$ by positive definiteness. ☐

Exercise 13.24. Prove the triangle inequality from Cauchy–Schwarz.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Expand $\|u + v\|^2$ and bound the real part of $\langle u, v\rangle$ by $\|u\|\|v\|$. ☐

Exercise 13.25. Give an example of two different inner products on $\mathbb{R}^2$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The standard product and $\langle x, y\rangle = 2x_1y_1 + x_2y_2$ are distinct inner products. ☐

Chapter summary

Inner products add geometry to linear algebra. Length, angle, orthogonality and best approximation all come from a positive definite sesquilinear form. The complex case forces careful placement of conjugation, but the resulting theory unifies Euclidean geometry with finite-dimensional Hilbert-space methods. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 14

Gram–Schmidt and Orthogonal Projection

Gram–Schmidt converts an arbitrary independent list into an orthonormal list spanning the same nested subspaces. Orthogonal projection then gives canonical decompositions and solves least-squares problems. This chapter connects exact geometry with the numerical algorithms QR factorization and normal equations.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use orthonormal lists in basis-free and coordinate form;
- explain and use gram–schmidt process in basis-free and coordinate form;
- explain and use projection onto a subspace in basis-free and coordinate form;
- explain and use best approximation in basis-free and coordinate form;
- explain and use qr factorization in basis-free and coordinate form;
- explain and use least squares in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

14.1 Orthonormal lists

A list is orthonormal when its vectors have norm one and are pairwise orthogonal. Coordinates in an orthonormal basis are inner products.

14.2 Gram–Schmidt process

Successively subtract from each vector its projections onto the previously constructed orthonormal vectors, then normalize.

14.3 Projection onto a subspace

For an orthonormal basis $e_1, \ldots, e_r$ of U, $P_U v = \sum_i \langle v, e_i \rangle e_i$.

14.4 Best approximation

$P_U v$ is the unique vector in U minimizing $\|v - u\|$.

14.5 QR factorization

Applying Gram–Schmidt to independent columns of A produces $A = QR$, with Q having orthonormal columns and R upper triangular.

14.6 Least squares

For full-column-rank A, the least-squares solution satisfies $A^*Ax = A^*b$ and can be computed more stably using QR.

14.7 Core structural results

Theorem 14.1 (Gram–Schmidt theorem)**.** *Every finite independent list in an inner-product space can be replaced by an orthonormal list with the same successive spans.*

Proof. At step k, subtract the projection of v_k onto the span already constructed. Independence guarantees the remainder is nonzero; normalization finishes the step. □

Theorem 14.2 (Projection theorem)**.** *For finite-dimensional U, every v decomposes uniquely as $v = P_U v + (v - P_U v)$ with the first term in U and the second in $U^{\perp}$.*

Proof. Use an orthonormal basis of U to define $P_U v$. Direct calculation shows the residual is orthogonal to every basis vector and hence to U. □

14.8 Worked examples

Example 14.3 (Two planar vectors)**.** Starting from $(1, 0)$ and $(1, 1)$, Gram–Schmidt returns $(1, 0)$ and $(0, 1)$.

Example 14.4 (Projection onto a line)**.** Projection onto the line spanned by unit u is $P(v) = \langle v, u \rangle u$.

Example 14.5 (Least squares line fitting)**.** The columns 1 and t form a design matrix; projection of data onto their column space gives the best affine fit.

14.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

14.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 14.6 (Gram–Schmidt by hand)**.** Apply Gram–Schmidt to $v_1 = (1, 1, 0)$ and $v_2 = (1, 0, 1)$.

Solution. Set $e_1 = v_1/\sqrt{2}$. The projection coefficient is $\langle v_2, e_1\rangle = 1/\sqrt{2}$, so $u_2 = v_2 - \frac{1}{2}(1, 1, 0) = (1/2, -1/2, 1)$. Its norm is $\sqrt{3/2}$, hence $e_2 = u_2/\sqrt{3/2}$. □

Problem 14.7 (Projection onto a line)**.** Project $v = (2, 3)$ onto the line spanned by $u = (1, 1)$.

Solution. Using the unit vector $e = u/\sqrt{2}$, $P(v) = \langle v, e\rangle e = (5/\sqrt{2})(1/\sqrt{2}, 1/\sqrt{2}) = (5/2, 5/2)$. □

Problem 14.8 (Projection matrix)**.** Find the projection matrix onto the line spanned by a unit vector u.

Solution. $P = uu^T$ in the real case. Indeed $Pv = u(u^T v)$ is exactly the scalar projection coefficient times u. □

Problem 14.9 (Best approximation)**.** Why is $P_U v$ the unique closest vector in U to v?

Solution. For $u \in U$, write $v - u = (v - P_U v) + (P_U v - u)$ with orthogonal summands. Pythagoras gives $\|v - u\|^2 = \|v - P_U v\|^2 + \|P_U v - u\|^2$, minimized uniquely at $u = P_U v$. □

Problem 14.10 (QR factorization)**.** What does Gram–Schmidt do to the columns of a full-rank matrix A?

Solution. It produces orthonormal columns Q and expresses each original column as a combination of the current and previous orthonormal columns. The coefficient matrix is upper triangular R, giving $A = QR$. □

Problem 14.11 (Normal equations)**.** Derive the least-squares normal equations.

Solution. At the minimizer, the residual $b - Ax$ is orthogonal to the column space of A. Therefore $A^*(b - Ax) = 0$, which rearranges to $A^*Ax = A^*b$. □

Problem 14.12 (Why QR is stabler)**.** Why is solving least squares through QR often preferable to normal equations?

Solution. Forming A^*A squares the spectral condition number, magnifying sensitivity. Orthogonal transformations used in QR have condition number 1 and preserve norms, so they avoid that amplification. □

Problem 14.13 (Projection idempotence)**.** Prove an orthogonal projection satisfies $P^2 = P$.

Solution. After one projection, Pv lies in the target subspace. Projecting a vector already in that subspace leaves it fixed, hence $P(Pv) = Pv$. ☐

Problem 14.14 (Complementary projection)**.** Show $I - P_U$ is the orthogonal projection onto $U^\perp$.

Solution. The orthogonal decomposition is $v = P_U v + (v - P_U v)$. The second term lies in $U^\perp$ and equals $(I - P_U)v$. Vectors already in $U^\perp$ are fixed. ☐

Problem 14.15 (Orthonormal coordinates)**.** If $e_1, \ldots, e_n$ is orthonormal, why is the coefficient of e_j in v equal to $\langle v, e_j \rangle$?

Solution. Write $v = \sum a_i e_i$ and inner-product with e_j. Orthonormality kills all terms except $a_j \langle e_j, e_j \rangle = a_j$. ☐

Problem 14.16 (Rank-deficient least squares)**.** What changes if the columns of A are dependent?

Solution. The least-squares projection onto the column space still exists uniquely, but the coefficient vector x need not be unique because nonzero kernel vectors can be added without changing Ax. The pseudoinverse selects the minimum-norm solution. ☐

Problem 14.17 (Projection synthesis)**.** Connect Gram–Schmidt, QR, and least squares in one sentence.

Solution. Gram–Schmidt constructs an orthonormal basis of the column space, QR records that construction algebraically, and least squares projects data onto that same column space using the orthonormal coordinates. ☐

14.11 Exercises with complete solutions

Exercise 14.18. Apply Gram–Schmidt to $(1, 1)$ and $(1, -1)$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. They are already orthogonal; normalize each by $\sqrt{2}$. ☐

Exercise 14.19. Project $(2, 3)$ onto the x-axis.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The projection is $(2, 0)$. ☐

Exercise 14.20. Why is the projection residual orthogonal to the subspace?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Subtracting all orthonormal-basis components makes its inner product with each basis vector zero. ☐

Exercise 14.21. Prove the best-approximation property.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. For $u \in U$, write $v - u = (v - P_U v) + (P_U v - u)$; the two terms are orthogonal, so Pythagoras is minimized exactly when $u = P_U v$. ☐

Exercise 14.22. What are the dimensions of Q and R for an $m \times n$ matrix of full column rank?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. In reduced QR, Q is $m \times n$ with orthonormal columns and R is invertible $n \times n$ upper triangular. ☐

Exercise 14.23. Why can normal equations be less stable than QR?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Forming A^*A squares the condition number, amplifying numerical sensitivity. ☐

Exercise 14.24. Show an orthogonal projection satisfies $P^2 = P$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Once a vector lies in U, projecting again leaves it unchanged; hence $P(Pv) = Pv$. ☐

Exercise 14.25. Show $I - P_U$ projects onto $U^\perp$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The decomposition theorem gives $v - P_U v \in U^\perp$, and vectors already in $U^\perp$ are fixed by $I - P_U$. ☐

Chapter summary

Gram–Schmidt converts an arbitrary independent list into an orthonormal list spanning the same nested subspaces. Orthogonal projection then gives canonical decompositions and solves least-squares problems. This chapter connects exact geometry with the numerical algorithms QR factorization and normal equations. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 15

Orthogonal and Unitary Operators

Orthogonal and unitary operators are the linear symmetries of inner-product geometry. They preserve lengths, angles and orthogonality, and their matrices satisfy $Q^*Q = I$. Reflections, rotations and Householder transformations show how geometric symmetries become structured matrices.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use isometries in basis-free and coordinate form;
- explain and use orthogonal and unitary matrices in basis-free and coordinate form;
- explain and use equivalent characterizations in basis-free and coordinate form;
- explain and use determinant and orientation in basis-free and coordinate form;
- explain and use reflections and householder maps in basis-free and coordinate form;
- explain and use rotations and planar blocks in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

15.1 Isometries

A linear map is an isometry when it preserves norms, equivalently inner products.

15.2 Orthogonal and unitary matrices

A real matrix is orthogonal when $Q^TQ = I$; a complex matrix is unitary when $Q^*Q = I$.

15.3 Equivalent characterizations

For a square matrix, orthonormal columns, orthonormal rows and inverse equal to adjoint are equivalent.

15.4 Determinant and orientation

A real orthogonal matrix has determinant ± 1. Determinant $+1$ gives orientation-preserving transformations.

15.5 Reflections and Householder maps

For unit u, $H = I - 2uu^*$ reflects across the hyperplane $u^\perp$.

15.6 Rotations and planar blocks

Real orthogonal operators decompose into invariant one- and two-dimensional pieces, with planar rotation blocks on the latter.

15.7 Core structural results

Theorem 15.1 (Inner-product preservation criterion)**.** *A linear map preserves inner products if and only if it preserves norms.*

Proof. One direction is immediate. For the converse, recover the inner product from the norm using the polarization identity. □

Theorem 15.2 (Adjoint inverse criterion)**.** *A matrix Q is unitary exactly when $Q^{-1} = Q^*$.*

Proof. This is simply a rearrangement of $Q^*Q = I$, with invertibility automatic. □

15.8 Worked examples

Example 15.3 (Planar rotation)**.** The standard rotation matrix is orthogonal with determinant 1.

Example 15.4 (Reflection)**.** $\operatorname{diag}(1, -1)$ reflects across the x-axis and has determinant -1.

Example 15.5 (Permutation matrix)**.** Every permutation matrix is orthogonal because its columns are a rearrangement of the standard orthonormal basis.

15.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

15.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 15.6 (Orthogonal inverse)**.** If $Q^TQ = I$, show $Q^{-1} = Q^T$.

Solution. The equation already says Q^T is a left inverse. For square matrices a left inverse is also a right inverse, so Q is invertible and $Q^{-1} = Q^T$. □

Problem 15.7 (Norm preservation)**.** Show an orthogonal matrix preserves Euclidean norm.

Solution. $\|Qx\|^2 = (Qx)^T(Qx) = x^TQ^TQx = x^Tx = \|x\|^2$. □

Problem 15.8 (Determinant sign)**.** Why does a real orthogonal matrix have determinant ±1?

Solution. Take determinants of $Q^TQ = I$: $(\det Q)^2 = 1$. Hence $\det Q = \pm1$. □

Problem 15.9 (Householder reflection)**.** For a unit vector u, show $H = I - 2uu^T$ is orthogonal.

Solution. Let $P = uu^T$, so $P^2 = P$ and $P^T = P$. Then $H^TH = H^2 = (I - 2P)^2 = I - 4P + 4P^2 = I$. □

Problem 15.10 (Reflection geometry)**.** What does $H = I - 2uu^T$ do to u and to vectors orthogonal to u?

Solution. $Hu = u - 2u = -u$. If $v \perp u$, then $u^Tv = 0$ and $Hv = v$. Thus H reflects across the hyperplane $u^\perp$. □

Problem 15.11 (Unitary eigenvalues)**.** Why must every eigenvalue of a unitary matrix have modulus 1?

Solution. If $Uv = \lambda v$ and $v \neq 0$, norm preservation gives $\|v\| = \|Uv\| = |\lambda|\|v\|$, so $|\lambda| = 1$. □

Problem 15.12 (Permutation matrices)**.** Why is every permutation matrix orthogonal?

Solution. Its columns are a reordering of the standard orthonormal basis, so they remain orthonormal. Hence $P^TP = I$. □

Problem 15.13 (Orientation)**.** What distinguishes rotations from reflections inside $O(n)$?

Solution. Both preserve inner products, but orientation-preserving orthogonal maps have determinant $+1$ and form $SO(n)$. Reflections have determinant -1. □

Problem 15.14 (Product of orthogonal matrices)**.** Show the product of two orthogonal matrices is orthogonal.

Solution. If $Q^TQ = I$ and $R^TR = I$, then $(QR)^T(QR) = R^TQ^TQR = R^TR = I$. □

Problem 15.15 (Columns and rows)**.** Why do orthonormal columns of a square matrix imply orthonormal rows?

Solution. Orthonormal columns give $Q^*Q = I$, so Q is invertible with inverse Q^*. Hence also $QQ^* = I$, which says the rows are orthonormal. □

Problem 15.16 (Condition number)**.** What is the Euclidean condition number of an orthogonal matrix?

Solution. All singular values equal 1, so $\kappa_2(Q) = 1$. Orthogonal maps neither stretch nor compress, making them optimally conditioned. ☐

Problem 15.17 (Symmetry synthesis)**.** Why are orthogonal/unitary transformations the natural coordinate changes for inner-product problems?

Solution. They preserve the inner product, so lengths, angles, orthogonality, numerical norms, and spectral geometry are unchanged. This lets one simplify coordinates without distorting the geometric structure under study. ☐

15.11 Exercises with complete solutions

Exercise 15.18. Show an orthogonal matrix preserves Euclidean norm.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. $\|Qx\|^2 = x^T Q^T Q x = x^T x$. ☐

Exercise 15.19. Why do orthogonal matrices have determinant ± 1?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Taking determinants in $Q^T Q = I$ gives $(\det Q)^2 = 1$. ☐

Exercise 15.20. Verify a Householder matrix is orthogonal.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. With $P = uu^T$ and $P^2 = P$, $(I - 2P)^2 = I$. Since it is symmetric, its transpose is its inverse. ☐

Exercise 15.21. What does a unitary operator do to an orthonormal basis?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. It sends it to another orthonormal basis because all pairwise inner products are preserved. ☐

Exercise 15.22. Show the inverse of a unitary matrix is unitary.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. If $Q^{-1} = Q^*$, then $(Q^{-1})^* = Q$ and $(Q^{-1})^* Q^{-1} = QQ^* = I$. ☐

Exercise 15.23. Give an orthogonal matrix with determinant -1 in $\mathbb{R}^3$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. For example $\mathrm{diag}(1, 1, -1)$. ☐

Exercise 15.24. Why is every eigenvalue of a complex unitary matrix of modulus one?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. If $Qv = \lambda v$, norm preservation gives $\|v\| = \|Qv\| = |\lambda|\|v\|$, so $|\lambda| = 1$. □

Exercise 15.25. Explain why orthogonal change of coordinates is numerically attractive.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Its inverse is the transpose and its condition number in the Euclidean norm is 1, so it does not amplify relative errors. □

Chapter summary

Orthogonal and unitary operators are the linear symmetries of inner-product geometry. They preserve lengths, angles and orthogonality, and their matrices satisfy $Q^*Q = I$. Reflections, rotations and Householder transformations show how geometric symmetries become structured matrices. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 16

The Spectral Theorem

The spectral theorem is the central structure theorem for self-adjoint, symmetric and normal operators. It says that the right compatibility with the inner product upgrades triangularization to orthonormal diagonalization. Functional calculus and positive operators then become simple operations on eigenvalues.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use adjoints in basis-free and coordinate form;
- explain and use self-adjoint and normal operators in basis-free and coordinate form;
- explain and use reality of the spectrum in basis-free and coordinate form;
- explain and use spectral theorem in basis-free and coordinate form;
- explain and use functional calculus in basis-free and coordinate form;
- explain and use positive operators in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

16.1 Adjoints

The adjoint T^* is characterized by $\langle Tv, w\rangle = \langle v, T^*w\rangle$.

16.2 Self-adjoint and normal operators

T is self-adjoint when $T = T^*$ and normal when $TT^* = T^*T$.

16.3 Reality of the spectrum

Eigenvalues of self-adjoint operators are real, and eigenspaces for distinct eigenvalues are orthogonal.

16.4 Spectral theorem

A finite-dimensional complex operator is normal if and only if it is unitarily diagonalizable. Over $\mathbb{R}$, symmetric operators are orthogonally diagonalizable.

16.5 Functional calculus

If $T = UDU^*$, define $f(T) = Uf(D)U^*$ for functions defined on the spectrum.

16.6 Positive operators

A self-adjoint operator is positive semidefinite when $\langle Tv, v\rangle \geq 0$; spectrally this means all eigenvalues are nonnegative.

16.7 Core structural results

Theorem 16.1 (Orthogonality of self-adjoint eigenspaces)**.** *Eigenvectors of a self-adjoint operator associated with distinct eigenvalues are orthogonal.*

Proof. If $Tv = \lambda v$ and $Tw = \mu w$, then $\lambda\langle v, w\rangle = \langle Tv, w\rangle = \langle v, Tw\rangle = \mu\langle v, w\rangle$. □

Theorem 16.2 (Finite-dimensional spectral theorem)**.** *Every self-adjoint operator on a finite-dimensional inner-product space has an orthonormal basis of eigenvectors.*

Proof. Choose an eigenvector, take its orthogonal complement, show that complement is invariant under self-adjointness, and induct on dimension. □

16.8 Worked examples

Example 16.3 (Real symmetric matrix)**.** $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ has orthogonal eigenvectors $(1, 1)$ and $(1, -1)$ with eigenvalues 3 and 1.

Example 16.4 (Positive square root)**.** For positive $A = UDU^*$, set $A^{1/2} = UD^{1/2}U^*$; then $A^{1/2}$ is positive and squares to A.

Example 16.5 (Normal but not self-adjoint)**.** A nontrivial complex unitary rotation is normal but may not be self-adjoint.

16.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;

3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

16.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 16.6 (Adjoint matrix)**.** In the standard complex inner product, what matrix represents the adjoint of A?

Solution. The adjoint is represented by the conjugate transpose $A^* = \overline{A}^T$, because $\langle Ax, y\rangle = x^*A^*y = \langle x, A^*y\rangle$. □

Problem 16.7 (Real eigenvalues)**.** Prove an eigenvalue of a self-adjoint operator is real.

Solution. If $Tv = \lambda v$, then $\lambda\|v\|^2 = \langle Tv, v\rangle = \langle v, Tv\rangle = \overline{\lambda}\|v\|^2$. Since $v \neq 0$, $\lambda = \overline{\lambda}$. □

Problem 16.8 (Orthogonal eigenspaces)**.** Why are eigenspaces of a self-adjoint operator for distinct eigenvalues orthogonal?

Solution. If $Tv = \lambda v$ and $Tw = \mu w$, self-adjointness gives $\lambda\langle v, w\rangle = \langle Tv, w\rangle = \langle v, Tw\rangle = \mu\langle v, w\rangle$. Distinctness forces the inner product to vanish. □

Problem 16.9 (Diagonalizing a symmetric matrix)**.** Diagonalize $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ orthogonally.

Solution. Eigenvalues are 3 and 1 with eigenvectors $(1, 1)$ and $(1, -1)$. Normalizing them gives an orthogonal matrix Q with $Q^TAQ = \mathrm{diag}(3, 1)$. □

Problem 16.10 (Positive square root)**.** How does the spectral theorem construct $A^{1/2}$ for positive semidefinite A?

Solution. Write $A = U\,\mathrm{diag}(\lambda_i)U^*$ with $\lambda_i \geq 0$. Define $A^{1/2} = U\,\mathrm{diag}(\sqrt{\lambda_i})U^*$. It is positive and squares to A. □

Problem 16.11 (Normal operator)**.** Why is every unitary operator normal?

Solution. For unitary U, $U^*U = UU^* = I$, so $UU^* = U^*U$. □

Problem 16.12 (Polynomial functional calculus)**.** If $T = UDU^*$ is normal, compute $p(T)$.

Solution. $p(T) = Up(D)U^*$, where $p(D)$ is obtained by applying p to each diagonal eigenvalue. This follows because powers of T preserve the same diagonalizing basis. □

Problem 16.13 (Positivity and spectrum)**.** Why is a self-adjoint operator positive semidefinite iff all eigenvalues are nonnegative?

Solution. In an orthonormal eigenbasis, $\langle Tv, v\rangle = \sum_i \lambda_i |v_i|^2$. This is nonnegative for all v exactly when each $\lambda_i \geq 0$. □

Problem 16.14 (Invariant orthogonal complement)**.** If U is invariant under self-adjoint T, show $U^\perp$ is invariant.

Solution. For $w \in U^\perp$ and $u \in U$, $\langle Tw, u\rangle = \langle w, Tu\rangle = 0$ because $Tu \in U$. Hence $Tw \in U^\perp$. □

Problem 16.15 (Spectral norm)**.** For self-adjoint T, what is $\|T\|_2$ in terms of eigenvalues?

Solution. Because the operator is unitarily diagonalizable, its singular values are $|\lambda_i|$. Therefore the operator norm is $\max_i |\lambda_i|$. ☐

Problem 16.16 (Spectral projector)**.** How can the projection onto an eigenspace be expressed when eigenvalues are distinct?

Solution. Use Lagrange interpolation: $P_i = \prod_{j\neq i}(T - \lambda_j I)/(\lambda_i - \lambda_j)$. It acts as 1 on the λ_i eigenspace and 0 on the others. ☐

Problem 16.17 (Spectral synthesis)**.** Why is the spectral theorem stronger than ordinary diagonalization?

Solution. It provides an orthonormal eigenbasis, not merely an arbitrary eigenbasis. Consequently the change-of-basis matrix is unitary, preserving the inner product and making geometric and numerical conclusions immediately available. ☐

16.11 Exercises with complete solutions

Exercise 16.18. Show eigenvalues of a self-adjoint operator are real.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. If $Tv = \lambda v$, then $\lambda\|v\|^2 = \langle Tv, v\rangle = \langle v, Tv\rangle = \overline{\lambda}\|v\|^2$. ☐

Exercise 16.19. Diagonalize $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ orthogonally.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Normalize eigenvectors $(1, 1)$ and $(1, -1)$ to form Q; then $Q^T AQ = \mathrm{diag}(3, 1)$. ☐

Exercise 16.20. Why is every unitary operator normal?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. $UU^* = U^*U = I$. ☐

Exercise 16.21. Show a positive semidefinite operator has nonnegative eigenvalues.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. For an eigenvector v, $\langle Tv, v\rangle = \lambda\|v\|^2 \geq 0$. ☐

Exercise 16.22. Define T^2 using spectral functional calculus.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. If $T = UDU^*$ then $T^2 = UD^2U^*$, corresponding to $f(t) = t^2$. ☐

Exercise 16.23. Why is the orthogonal complement of an invariant eigenspace also invariant for a self-adjoint operator?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. If w is orthogonal to the eigenspace and v is in it, then $\langle Tw, v\rangle = \langle w, Tv\rangle = \lambda\langle w, v\rangle = 0$. ☐

Exercise 16.24. What extra conclusion does normality give beyond triangularization?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. A normal operator can be diagonalized by a unitary basis change, so the basis can be chosen orthonormal. ☐

Exercise 16.25. Relate spectral theorem to quadratic forms.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. A real symmetric matrix can be orthogonally diagonalized, reducing its quadratic form to a sum of scalar multiples of squares. ☐

Chapter summary

The spectral theorem is the central structure theorem for self-adjoint, symmetric and normal operators. It says that the right compatibility with the inner product upgrades triangularization to orthonormal diagonalization. Functional calculus and positive operators then become simple operations on eigenvalues. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 17

Quadratic Forms

Quadratic forms encode second-order geometry. Symmetric bilinear forms and symmetric matrices represent the same finite-dimensional data once a basis is chosen. Congruence, not similarity, is the natural basis-change relation, and Sylvester's law of inertia identifies the basis-independent signature.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use quadratic and bilinear forms in basis-free and coordinate form;
- explain and use matrix representation in basis-free and coordinate form;
- explain and use polarization in basis-free and coordinate form;
- explain and use congruence in basis-free and coordinate form;
- explain and use sylvester law of inertia in basis-free and coordinate form;
- explain and use definiteness in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

17.1 Quadratic and bilinear forms

A quadratic form on a real vector space has the form $q(v) = B(v, v)$ for a symmetric bilinear form B.

17.2 Matrix representation

In coordinates, $q(x) = x^T Ax$ with A symmetric. Cross terms appear twice the corresponding off-diagonal entry.

17.3 Polarization

Over characteristic not 2, the symmetric bilinear form is recovered from q by $B(u,v) = \frac{1}{2}(q(u+v) - q(u) - q(v))$.

17.4 Congruence

Under coordinate change $x = Py$, the matrix transforms as $A \mapsto P^T AP$.

17.5 Sylvester law of inertia

Every real symmetric quadratic form is congruent to a diagonal form with entries $1, -1, 0$, and the counts of positive, negative and zero entries are invariant.

17.6 Definiteness

A symmetric form is positive definite exactly when all eigenvalues are positive; equivalent criteria include positivity of leading principal minors.

17.7 Core structural results

Theorem 17.1 (Polarization identity)**.** *For a real quadratic form $q(v) = B(v,v)$ with symmetric bilinear B, $B(u,v) = \frac{1}{2}(q(u+v) - q(u) - q(v))$.*

Proof. Expand $q(u+v) = B(u,u) + 2B(u,v) + B(v,v)$ and rearrange. ☐

Theorem 17.2 (Sylvester law of inertia)**.** *The numbers of positive, negative and zero squares in a real diagonalization by congruence are basis-independent.*

Proof. A congruence change is an invertible change of variables; maximal dimensions of subspaces on which the form is positive or negative definite are intrinsic and determine the counts. ☐

17.8 Worked examples

Example 17.3 (Conic form)**.** $x^2+2xy+y^2 = (x+y)^2$ has rank one and is positive semidefinite.

Example 17.4 (Indefinite form)**.** $x^2 - y^2$ has inertia $(1,1,0)$ and vanishes on the two lines $x = \pm y$.

Example 17.5 (Hessian viewpoint)**.** At a critical point of a smooth function, the Hessian defines a quadratic form whose signature describes local second-order behavior.

17.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;
3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

17.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 17.6 (Matrix of a quadratic form)**.** Find the symmetric matrix for $q(x, y) = 3x^2 + 4xy + 2y^2$.

Solution. $q = [x\ y] \begin{pmatrix} 3 & 2 \\ 2 & 2 \end{pmatrix} \binom{x}{y}$ because the cross term is $2a_{12}xy = 4xy$. □

Problem 17.7 (Polarization)**.** Recover the symmetric bilinear form B from a real quadratic form q.

Solution. $B(u, v) = \frac{1}{2}(q(u+v)-q(u)-q(v))$. Expanding $q(u+v) = B(u, u)+2B(u, v)+B(v, v)$ proves the formula. □

Problem 17.8 (Congruence versus similarity)**.** Why does a basis change send a quadratic-form matrix to P^TAP?

Solution. If old coordinates are $x = Py$, then $q = x^TAx = y^TP^TAPy$. Both copies of the coordinate vector change, producing congruence rather than similarity. □

Problem 17.9 (Definiteness by eigenvalues)**.** How can the spectral theorem test positive definiteness of a real symmetric matrix?

Solution. Orthogonally diagonalize $A = QDQ^T$. Then $x^TAx = y^TDy = \sum_i \lambda_i y_i^2$. This is positive for all nonzero x exactly when every eigenvalue is positive. □

Problem 17.10 (Indefinite form)**.** Classify $q(x, y) = x^2 - y^2$.

Solution. It takes positive values, such as at $(1, 0)$, and negative values, such as at $(0, 1)$, so it is indefinite. Its inertia is $(1, 1, 0)$. □

Problem 17.11 (Rank-one semidefinite form)**.** Classify $q(x, y) = (x + y)^2$.

Solution. The form is nonnegative and vanishes on the nonzero line $x+y = 0$, so it is positive semidefinite but not definite. Its matrix has rank 1. □

Problem 17.12 (Diagonalizing a cross term)**.** Diagonalize $2xy$ by an orthogonal change of variables.

Solution. Set $u = (x + y)/\sqrt{2}$, $v = (x - y)/\sqrt{2}$. Then $x = (u + v)/\sqrt{2}$, $y = (u - v)/\sqrt{2}$, so $2xy = u^2 - v^2$. □

Problem 17.13 (Inertia invariance)**.** What intrinsic meaning do the positive and negative inertia indices have?

Solution. They are the maximal dimensions of subspaces on which the quadratic form is positive definite or negative definite. Because those maximal dimensions are basis-independent, the inertia is invariant under congruence. □

Problem 17.14 (Hessian interpretation)**.** At a critical point, how does the Hessian quadratic form classify a nondegenerate point?

Solution. Positive definite Hessian gives a strict local minimum, negative definite gives a strict local maximum, and an indefinite Hessian gives a saddle point. The signature captures the second-order geometry. □

Problem 17.15 (Principal-minor criterion)**.** For $A = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, state the positive-definite criterion using leading principal minors.

Solution. Sylvester's criterion gives $a > 0$ and $ac - b^2 > 0$. These conditions are equivalent to both eigenvalues being positive. ☐

Problem 17.16 (Null directions)**.** What does a zero eigenvalue of a symmetric quadratic-form matrix mean geometrically?

Solution. It gives a nonzero direction v with $Av = 0$, hence $q(v) = 0$ and no quadratic curvature in that direction. The form is degenerate. ☐

Problem 17.17 (Quadratic-form synthesis)**.** Why is signature more appropriate than individual matrix entries as geometric data?

Solution. Entries depend strongly on coordinates, while signature is invariant under invertible changes of variables. It records the intrinsic counts of positive, negative, and null directions. ☐

17.11 Exercises with complete solutions

Exercise 17.18. Find the symmetric matrix for $q(x, y) = 2x^2 + 6xy + 5y^2$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. $A = \begin{pmatrix} 2 & 3 \\ 3 & 5 \end{pmatrix}$ because the cross term is $2a_{12}xy$. ☐

Exercise 17.19. Recover the bilinear form from a quadratic form by polarization.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Use $B(u, v) = \frac{1}{2}(q(u + v) - q(u) - q(v))$. ☐

Exercise 17.20. Why does a basis change act by congruence rather than similarity?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Both copies of the coordinate vector change in $x^T Ax$, producing $y^T P^T APy$. ☐

Exercise 17.21. Determine the definiteness of $x^2 - y^2$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. It is indefinite because it takes both positive and negative values. ☐

Exercise 17.22. What is the inertia of $x^2 + y^2$?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. $(2, 0, 0)$: two positive directions, no negative or null directions. ☐

Exercise 17.23. Show positive definite implies nonsingular.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. A nonzero kernel vector would satisfy $q(v) = v^T Av = 0$, contradicting positive definiteness. ☐

Exercise 17.24. Diagonalize $2xy$ by a linear change of variables.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Set $u = (x + y)/\sqrt{2}$, $v = (x - y)/\sqrt{2}$; then $2xy = u^2 - v^2$. ☐

Exercise 17.25. Explain the role of the spectral theorem for symmetric quadratic forms.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Orthogonal diagonalization of the symmetric matrix expresses the form as a weighted sum of coordinate squares. ☐

Chapter summary

Quadratic forms encode second-order geometry. Symmetric bilinear forms and symmetric matrices represent the same finite-dimensional data once a basis is chosen. Congruence, not similarity, is the natural basis-change relation, and Sylvester's law of inertia identifies the basis-independent signature. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.

Chapter 18

Singular-Value Decomposition

The singular-value decomposition is the universal orthogonal factorization of a linear map between finite-dimensional inner-product spaces. Unlike eigenvalue decomposition, it applies to every rectangular matrix. Singular values quantify stretching, rank, conditioning and optimal low-rank approximation.

Learning goals

By the end of the chapter, the reader should be able to:

- explain and use singular values in basis-free and coordinate form;
- explain and use svd theorem in basis-free and coordinate form;
- explain and use construction from the spectral theorem in basis-free and coordinate form;
- explain and use rank and norms in basis-free and coordinate form;
- explain and use pseudoinverse in basis-free and coordinate form;
- explain and use low-rank approximation in basis-free and coordinate form;

Conceptual roadmap

The chapter is organized around a repeated pattern:

$$\boxed{\text{definition} \longrightarrow \text{structural theorem} \longrightarrow \text{coordinate calculation} \longrightarrow \text{application.}}$$

Whenever possible, we separate the invariant mathematical object from a chosen coordinate representation.

18.1 Singular values

The singular values of A are the nonnegative square roots of the eigenvalues of A^*A.

18.2 SVD theorem

Every $m \times n$ matrix has a factorization $A = U\Sigma V^*$ with U, V unitary (orthogonal over $\mathbb{R}$) and Σ diagonal rectangular with nonnegative entries.

18.3 Construction from the spectral theorem

Diagonalize A^*A to obtain right singular vectors; normalize their images under A to obtain left singular vectors.

18.4 Rank and norms

The rank is the number of positive singular values. The operator norm is the largest singular value, and the Frobenius norm squared is the sum of squared singular values.

18.5 Pseudoinverse

Replacing each nonzero singular value by its reciprocal gives the Moore–Penrose pseudoinverse.

18.6 Low-rank approximation

Truncating the SVD after the largest r singular values gives a best rank-r approximation in operator and Frobenius norms.

18.7 Core structural results

Theorem 18.1 (Singular-value decomposition)**.** *For every matrix $A \in \mathbb{F}^{m\times n}$, there exist unitary U, V and a rectangular diagonal Σ with nonnegative entries such that $A = U\Sigma V^*$.*

Proof. Apply the spectral theorem to the positive semidefinite matrix A^*A. For each positive eigenvalue σ_i^2, set $u_i = Av_i/\sigma_i$; extend the resulting orthonormal lists to bases. □

Theorem 18.2 (Rank from singular values)**.** *The number of nonzero singular values equals* rank A.

Proof. The kernels of A and A^*A coincide, so they have the same nullity. Rank–nullity then shows A^*A has as many positive eigenvalues as the rank of A. □

18.8 Worked examples

Example 18.3 (Diagonal matrix)**.** For a real diagonal matrix, singular values are the absolute values of its diagonal entries.

Example 18.4 (Rank-one matrix)**.** If $A = uv^*$, there is a single nonzero singular value $\|u\|\|v\|$.

Example 18.5 (Condition number)**.** For invertible A, the Euclidean condition number is $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$.

18.9 Diagnostic checklist

Before moving on, it is useful to distinguish four levels of understanding:

1. recognize the definitions in unfamiliar examples;
2. prove the core structural statements without coordinates when possible;

3. translate the same statement into a matrix or coordinate computation;
4. know which hypotheses are essential and produce a counterexample when one is removed.

18.10 Solved dossiers

These dossiers are longer worked problems. They complement the shorter exercise–hint–solution triads by combining several ideas in one argument.

Problem 18.6 (Singular values of a diagonal matrix)**.** Find the singular values of $\operatorname{diag}(3, -2, 0)$.

Solution. $A^T A = \operatorname{diag}(9, 4, 0)$, so the singular values are the nonnegative square roots $3, 2, 0$. ☐

Problem 18.7 (Kernel identity)**.** Prove $\ker(A^*A) = \ker A$.

Solution. If $Av = 0$, then $A^*Av = 0$. Conversely, if $A^*Av = 0$, then $0 = \langle A^*Av, v\rangle = \langle Av, Av\rangle = \|Av\|^2$, so $Av = 0$. ☐

Problem 18.8 (Rank from singular values)**.** Why does the number of nonzero singular values equal $\operatorname{rank} A$?

Solution. A^*A has the same kernel as A, hence the same nullity. Rank-nullity then implies the number of positive eigenvalues of A^*A, i.e. nonzero singular values, equals the rank of A. ☐

Problem 18.9 (Rank-one SVD)**.** If $A = uv^*$ with nonzero u, v, what is its nonzero singular value?

Solution. Write $A = (u/\|u\|)(\|u\|\|v\|)(v/\|v\|)^*$. This is already a rank-one SVD, so the sole nonzero singular value is $\|u\|\|v\|$. ☐

Problem 18.10 (Operator norm)**.** Why is $\|A\|_2$ the largest singular value?

Solution. In an SVD $A = U\Sigma V^*$, unitary factors preserve norm. Thus maximizing $\|Ax\|$ over unit x reduces to maximizing $\|\Sigma y\|$, whose largest stretch is the largest diagonal singular value. ☐

Problem 18.11 (Frobenius norm)**.** Express $\|A\|_F^2$ in terms of singular values.

Solution. Unitary invariance of the Frobenius norm gives $\|A\|_F^2 = \|\Sigma\|_F^2 = \sum_i \sigma_i^2$. ☐

Problem 18.12 (Pseudoinverse)**.** How is the Moore–Penrose pseudoinverse constructed from an SVD?

Solution. If $A = U\Sigma V^*$, define $A^+ = V\Sigma^+U^*$, where each nonzero singular value σ_i is replaced by $1/\sigma_i$ and zeros remain zero. ☐

Problem 18.13 (Condition number)**.** For invertible A, why is $\kappa_2(A) = \sigma_{\max}/\sigma_{\min}$?

Solution. $\|A\|_2 = \sigma_{\max}$ and $\|A^{-1}\|_2 = 1/\sigma_{\min}$. Their product is the stated ratio. ☐

Problem 18.14 (Rectangular geometry)**.** Why does the SVD use two different orthonormal bases?

Solution. A rectangular map has different domain and codomain spaces. Right singular vectors form an orthonormal basis of input directions; left singular vectors form orthonormal output directions. The diagonal Σ records the stretch between matched directions. ☐

Problem 18.15 (Low-rank truncation)**.** What is the rank of the matrix obtained by keeping only the first r nonzero SVD terms?

Solution. Each term $\sigma_i u_i v_i^*$ has rank one and the corresponding left/right vectors are orthogonal. Keeping r positive terms produces rank exactly r. ☐

Problem 18.16 (Best approximation principle)**.** State the Eckart–Young conclusion for truncated SVD.

Solution. The matrix obtained by keeping the largest r singular values is a best rank-r approximation in both spectral and Frobenius norms. The spectral error is σ_{r+1} and the squared Frobenius error is $\sum_{i>r} \sigma_i^2$. ☐

Problem 18.17 (SVD synthesis)**.** How does SVD unify rank, conditioning, least squares, and data compression?

Solution. Nonzero singular values determine rank; their ratio controls conditioning; the pseudoinverse solves least-squares/minimum-norm problems; and decay of singular values quantifies how accurately the map can be compressed to low rank. One factorization exposes all four phenomena. ☐

18.11 Exercises with complete solutions

Exercise 18.18. Find the singular values of $\operatorname{diag}(3, -2)$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. They are 3 and 2, the square roots of eigenvalues 9 and 4 of $A^T A$. ☐

Exercise 18.19. Why is A^*A positive semidefinite?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. $\langle A^*Av, v\rangle = \langle Av, Av\rangle = \|Av\|^2 \geq 0$. ☐

Exercise 18.20. Show $\ker(A^*A) = \ker A$.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. If $Av = 0$ then clearly $A^*Av = 0$. Conversely, $A^*Av = 0$ implies $0 = \langle A^*Av, v\rangle = \|Av\|^2$. ☐

Exercise 18.21. What is the rank of a matrix with singular values $5, 2, 0, 0$?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. The rank is 2, the number of nonzero singular values. ☐

Exercise 18.22. Describe the pseudoinverse of a diagonal singular-value matrix.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. ☐

Solution. Transpose the rectangular shape and replace each positive σ_i by $1/\sigma_i$, leaving zeros unchanged. □

Exercise 18.23. Why is SVD available for rectangular matrices while diagonalization is not?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. SVD uses the square self-adjoint matrices A^*A and AA^* and two possibly different orthonormal bases; ordinary diagonalization requires an endomorphism. □

Exercise 18.24. What does the largest singular value measure?

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. It is $\max_{\|x\|=1} \|Ax\|$, the maximum stretching factor. □

Exercise 18.25. Explain truncated SVD conceptually.

Hint. Identify the definition or structural theorem in this chapter that directly controls the question, then reduce the calculation to that statement. □

Solution. Keep only the dominant orthogonal input/output modes and their singular values; the discarded modes contribute the smallest possible approximation error for the prescribed rank. □

Chapter summary

The singular-value decomposition is the universal orthogonal factorization of a linear map between finite-dimensional inner-product spaces. Unlike eigenvalue decomposition, it applies to every rectangular matrix. Singular values quantify stretching, rank, conditioning and optimal low-rank approximation. The important habit is to move fluently between invariant statements and concrete coordinates while remembering which conclusions depend on finite dimensionality, the scalar field, or the inner product.